WAGER: Within-cell Antisymmetric Gain Evaluation of Resolution

Quang-Vinh Dang[a,*]

[a] British University Vietnam, Ecopark Township, Hung Yen, Vietnam

Abstract

Comparing two probabilistic predictors on data whose labels are partly determined by a known nuisance feature raises an attribution question aggregate scores cannot answer: when one model outscores another, how much of the gain reflects a closer fit to the nuisance distribution rather than better prediction of individual cases? We introduce WAGER, an estimator that answers it directly. For two frozen models it forms their proper-score contrast, matches test cases sharing a declared discrete feature, and crosses their labels, giving an exact finite-sample identity: the observed gain equals a prior-transported gain plus an antisymmetric residual. The residual is an order-two U-statistic equal, under the quadratic score, to the within-cell prediction–label covariance the new model has gained, identifying it with the resolution term of the classical proper-score decomposition applied to a model pair rather than a single forecaster. The estimator costs $O(NK)$, fits no nuisance model, has no tuning parameter, and admits Hájek influence-function inference with cluster-robust intervals and a stratified randomization test. Two exact results follow: leave-one-out transport removes an $(n_c - 1)/n_c$ attenuation that the in-sample plug-in incurs, and coarsening the matching feature shifts the estimand by an explicit between-cell covariance. The same covariance identity is shown to hold for the entire Bregman-score family, an asymptotic normality theorem justifies the reported intervals in the large-sample limit and reduces, at a trivial matching feature, to a clustered Diebold–Mariano test, and a Cauchy–Schwarz sensitivity bound quantifies how far an unrecorded confounder could move the residual away from what a fully adjusted audit would find. Simulation establishes interval coverage and discriminant validity, including the residual's sensitivity to monotone recalibration and the matched protocol that removes it. Applied to scene-graph prediction, long-tailed image classification, and long-tailed text classification, the decomposition shows published methods whose headline improvements are almost entirely nuisance-transported while instance-level alignment is statistically unchanged.

Keywords: Model comparison, Proper scoring rules, U-statistics, Model assessment, Pattern recognition, Randomization tests

*Corresponding author.
Email address: `vinh.dq4@buv.edu.vn` (Quang-Vinh Dang)

1. Introduction

Model comparison on a benchmark is an estimation problem that is usually treated as a lookup. Two predictors are scored, the difference is reported, and the difference is read as evidence that one model has learned something the other has not. That reading is sound only when the labels are not already partly determined by a feature both models observe. When they are — and in many benchmarks they conspicuously are — a model can raise its score by fitting the conditional label distribution of that nuisance feature more closely, without predicting any individual case better. The score difference then confounds two quantities that a careful evaluation should separate, and no amount of significance testing on the difference itself will separate them.

Computer-vision benchmarks make the problem concrete, and supply the applications of this paper. In scene graph generation (SGG), the ordered object-class pair strongly predicts the relation label; in visual question answering (VQA), question form predicts the answer; in long-tail recognition, category frequency dominates aggregate performance. Models can therefore improve a headline score by fitting annotation regularities more precisely without improving the instance-specific association between an image and its label. This is not merely hypothetical: frequency baselines are competitive on Visual Genome [59], VQA systems fail under changed answer priors [20, 1], standard metrics obscure tail behavior [35], and the same diagnosis continues to reappear in current work: open-vocabulary and knowledge-enhanced SGG methods report aggregate scores – increasingly sliced by predicate frequency, but with the composition of any given model-pair improvement left unattributed [34, 60, 48], long-tailed recognition methods rebalance training toward rare classes without attributing a resulting gain to the frequency prior versus instance-specific evidence [39, 33, 38], and recent benchmark audits show multimodal models can exploit non-visual shortcuts that inflate exactly this kind of aggregate score [5, 62].

The usual audit trains or constructs a prior-only baseline and compares its score with a full model. That strategy is useful diagnostically, but it does not identify where the *gain between two submitted systems* came from. Baseline estimation error, smoothing, calibration, and model-family mismatch all enter the result. More fundamentally, separately decomposing each model and subtracting the decompositions is an indirect answer to a pairwise question.

We instead ask a counterfactual question at the level of the gain itself. Let f_0 be the old model, f_1 the claimed improvement, and $\phi(x)$ the declared prior feature. For two test examples i, j with the same ϕ, we retain both models' prediction vectors but cross the observed labels. The crossed assignment preserves the prior cell and its label histogram while destroying which prediction vector belongs with which label. Comparing the observed and crossed assignments measures whether $f_1 - f_0$ improves this instance-level alignment. Figure 1 summarizes the idea.

This construction gives WAGER: *Within-cell Antisymmetric Gain Evaluation of Resolution*. The name is deliberately literal. What the residual measures is a resolution difference in the sense of the classical proper-score literature [41, 4] — predictive alignment with the correct instance beyond ϕ — and we call it the instance-alignment gain throughout, with no claim about cognition, causality, or understanding attached. WAGER makes nine contributions.

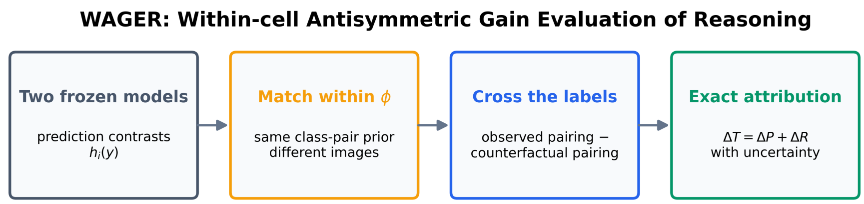

Figure 1: **WAGER audits the gain directly, on the benchmark's own examples.** Two VG150 test relations share the audit cell ϕ = (man, surfboard) yet carry different labels. Crossing the labels within the cell (arrows) preserves the cell and its label histogram while destroying which prediction meets which image. Scoring two frozen models under the observed and transported assignments splits their score difference exactly into a prior-transported gain and an instance-alignment gain.

1. We define a direct model-pair estimand whose prior component is generated by within-cell label transport, not by fitting a frequency forecaster.
2. We prove an exact population and finite-sample decomposition. The residual is an antisymmetric U-statistic; under the quadratic score it has a covariance identity that identifies it with the resolution term of the classical proper-score decomposition [41, 12, 4], applied to a model-pair contrast rather than to one forecaster.
3. We prove two further exact results that were not available for the single-model decomposition: leave-one-out transport removes a finite-sample attenuation bias of $(n_c - 1)/n_c$ that a naive in-sample plug-in prior would otherwise incur at every finite cell size (Proposition 3), and coarsening the audit cell changes the estimate by an explicit between-cell covariance term (Proposition 1), which turns an empirical sensitivity finding into a population-level mechanism.
4. We generalize the covariance identity from the quadratic score to the entire Bregman-score family (Theorem 2), recovering the quadratic and log-score identities as two named cases of one result, and we give a Cauchy–Schwarz/Popoviciu sensitivity bound (Proposition 2) that converts the paper's central qualitative caveat – an unrecorded confounder can masquerade as alignment – into a numeric limit on how far the reported residual could be from a fully adjusted audit.
5. We derive an $O(NK)$ algorithm, a Hájek influence function for image-cluster-robust intervals, and a cell-stratified randomization test under a sharp exchangeability null.
6. We prove an asymptotic normality theorem for the dataset-level estimator (Theorem 4) that justifies its reported intervals in the large-sample limit, and show as a corollary that, at a trivial audit feature, the undecomposed statistic and interval are exactly a clustered Diebold–Mariano/Giacomini–White test [13, 18] for the paired loss differential (Corollary 3), situating WAGER precisely relative to comparative forecast evaluation rather than only by analogy.
7. We validate the estimator under controlled alternatives and on Visual Genome, where it distinguishes a class-only gain from gains created by adding spatial evi-

dence, and further from gains created by a model that consumes real image pixels through a frozen CLIP encoder rather than only box geometry (Section 4.6).

8. We audit a published debiasing method rather than only our own predictors: applied to two released scene-graph checkpoints on the standard benchmark, WAGER shows that a ten-point mean-recall improvement is accompanied by no measurable change in within-cell instance alignment, its whole proper-score difference lying in the prior channel (Section 4.5).
9. We show WAGER is not SGG-specific by applying it unchanged to two further domains and task types – long-tailed image classification on CIFAR-100-LT (Section 4.7) and long-tailed text classification on 20-Newsgroups-LT (Section 4.8) – decomposing the gain of class-balanced and deferred re-weighting classifiers over a vanilla cross-entropy baseline into class-frequency-prior and instance-alignment components in both a vision and a language task, giving empirical support to the broader-applicability claim of Section 5.2 rather than leaving it as an untested hypothesis.

WAGER is also defined by what it avoids: it fits no prior baseline model, requires no projection/evaluation split, and imposes no decision threshold on a gain ratio. The primary quantity is the signed instance-alignment gain with uncertainty. Its scope is correspondingly concrete: WAGER audits closed-set probabilistic prediction, requiring both models' full probability vectors and a discrete, declarable prior feature. Any benchmark meeting those three conditions is in reach; generative or free-form evaluations, where no K-way probability vector exists, are not, and we do not claim them.

2. Related Work

2.1. Benchmark priors and shortcut learning

Dataset bias has long complicated comparisons between visual recognition systems [52, 17]. In SGG, the FREQ predictor estimates $p(\text{predicate} \mid \text{subject}, \text{object})$ and is surprisingly competitive [59]; mean recall and causal interventions were proposed to reduce this dependence [9, 50, 49], although metric pathologies remain [35]. More recent SGG work continues this line under open-vocabulary and knowledge-enhanced settings [34, 60, 48]. The community's reporting standards have themselves responded to the shortcut: mean recall is reported alongside recall, often broken down by predicate-frequency group [49, 35], and zero-shot recall – introduced with the earliest language-prior models [37] – checks whether a gain survives on triplets absent from training, exactly the frequency-exploitation worry [49]. In VQA, GQA-OOD stratifies evaluation by answer rarity within question groups [28], and VQA-CP changes language priors between train and test [1], though changing-prior splits can themselves be exploited [51]; recent VQA work continues to target the same language-prior shortcut [57, 31]. Benchmark-level audits of exploitable non-visual shortcuts have also expanded to broader multimodal evaluation [5, 62]. All of these are prior-aware slicings, splits, or stress tests of a *single* model's performance, and they are effective as far as they go. What none of them provides is what WAGER targets: an exact decomposition, with uncertainty, of the score difference between two submitted systems into a prior-transported channel and an instance-alignment channel. A stratified or zero-shot metric can show *that* a gain is uneven across the prior; it does not attribute the pairwise gain itself, on the full evaluation set, into the two channels with a finite-sample identity and inference.

2.2. Long-tailed recognition

The same shortcut, in a structurally identical form, appears well outside scene graph generation. A long-tailed classifier can raise its aggregate accuracy simply by fitting the training-time class-frequency prior more precisely, without becoming any better at discriminating instances within a class [39, 14]. The canonical remedies rebalance the training objective or the classifier itself: effective-number class-balanced re-weighting [11], deferred re-weighting [7], logit adjustment [40], and post-hoc re-training or re-scaling of a decoupled classifier [27], with a continuing stream of recent variants [39, 33, 38], much as mean-recall and causal-intervention methods rebalance SGG training. What such remedies share with their SGG counterparts is that they intervene on how a model is trained or adjusted rather than on how a trained pair is compared, so none of them attributes an observed score gap between two finished models to frequency-fitting on the one hand and instance-specific improvement on the other. That is the question Section 4.7 puts to WAGER, using the benchmark's own coarse superclasses as the declared prior feature ϕ; the estimator exposes the composition of two re-weighting schedules' gains that their aggregate scores leave hidden – one whose near-zero total conceals a large prior improvement cancelling an equally large alignment loss, the other whose calibration-matched improvement is almost entirely rare-class instance alignment.

2.3. Information and predictive probing

Usable-information analyses measure information accessible to a chosen predictor family [58, 15, 24]. Their central objects are model- or family-level quantities. WAGER is deliberately narrower and more operational: it attributes the proper-score difference between two concrete frozen models. We do not identify conditional mutual information and do not claim that a positive residual establishes causal or compositional understanding.

2.4. Proper scores, randomization, and U-statistics

Strictly proper scoring rules reward honest predictive distributions [19]; WAGER uses their model-pair contrast. A recent survey revisits estimation and forecast evaluation under proper scoring rules broadly [55], and the reliability–resolution–uncertainty partition itself continues to be re-derived and extended: to general strictly proper scores and multiclass outcomes [8], to entropy/divergence-based uncertainty quantification [26], and via an alternative covariance rearrangement for the Brier score specifically [54]. WAGER uses the same partition but applies it to a paired gain contrast rather than to one forecaster, as detailed in Section 2.5. Permutation methods have been used to assess classifier performance and conditional independence [42, 3], with recent work characterizing when generalized permutation tests attain optimal power [29] and extending conditional-independence testing via ball-divergence measures [2] and transport-map reductions to the unconditional case [21]. Conditional permutation tests typically estimate or assume a conditional distribution when confounders are continuous or high-dimensional. WAGER exploits a different setting: the benchmark supplies a discrete audit cell ϕ, and the target is not generic conditional independence but a specific model-to-model score gain. This permits exact within-cell transport without estimating a nuisance law. The antisymmetric residual is an order-two U-statistic in the sense of Hoeffding [25]; its Hájek-projection inference is part of the same broader program of finite- and large-sample U-statistic theory that recent work extends to anytime-valid confidence sequences [6], enabling an efficient all-pairs estimator and influence-function inference here. Pairwise betting has also been used to test exchangeability sequentially [46]; WAGER is a fixed-sample gain decomposition and makes no sequential-validity claim – the sequential variant a live leaderboard would need, with anytime validity as submissions arrive, is a distinct open problem. Its new element is the gain-contrast transport identity and the resulting attribution algorithm, rather than permutation or U-statistic theory by itself.

Three adjacent literatures locate the same object from different directions. Inference on a paired score differential is the classical territory of comparative forecast evaluation: the Diebold–Mariano test [13] and conditional predictive ability tests [18] ask whether one forecaster beats another, optionally conditional on covariates – the closest existing relative of a gain evaluated conditional on ϕ. WAGER adds to that tradition an exact decomposition of the differential itself, and Corollary 3 makes the relationship precise: at the trivial partition, WAGER's own total-gain statistic and its image-clustered interval literally are a grouped Diebold–Mariano statistic, so the decomposition is best read as everything DM/GW already deliver plus the further split that neither provides. Second, the prior channel is precisely the component that label-shift procedures manipulate post hoc: re-weighting a classifier's outputs to a new label distribution [45, 36] can add or remove prior-transported gain without touching instance-level evidence, which is what the prior-matching experiment of Section 4.6 exploits as a falsification test. Third, evaluating calibration within cells of a declared grouping, uniformly over many candidate groupings, is the subject of multicalibration [23]; that literature is the formal treatment of the multiplicity question raised by the choice of ϕ (Section 5). A fourth, more classical connection concerns what an unrecorded grouping variable can hide: omitted-variable sensitivity analysis in observational causal inference [44] and its recent correlation- and moment-parameterized refinements [16, 61] bound how much an unmeasured confounder could change a causal estimate; Proposition 2 adapts the same Cauchy–Schwarz style of argument to bound how much an unrecorded covariate could change the reported alignment gain, which is the same question – how far can what is unobserved move the estimate – asked of a different estimand.

2.5. Relation to classical proper-score decompositions

The covariance identity in Eq. (7) is a relative of the reliability resolution–uncertainty partition of a single proper score [41, 12], later generalized and made conditioning-set explicit by Bröcker [4]: conditioned on a chosen grouping variable, a proper score splits into a calibration term and a resolution term, and the resolution term is a covariance between the forecast and the outcome indicator within groups. We deliberately build on this fact rather than obscure it. WAGER is not a new proper score, and ΔR_c is best understood as a *resolution difference*, not a new quantity to be evaluated on its own. Two things distinguish WAGER from applying the classical decomposition to q_0 and q_1 separately and subtracting the two resolutions. First, that indirect route requires two resolution estimates, each of which is generally biased in finite samples by the same self-prediction mechanism formalized in Section 3.8; the bias does not cancel under subtraction because it scales with each model's own cell counts. WAGER instead forms the paired gain vector $H_x(y)$ first and decomposes the *contrast* in one pass, which is what makes the exact finite-sample identity and the antisymmetric U-statistic kernel possible. Second, the classical decomposition is a diagnostic for a single forecaster's calibration and sharpness; it neither targets nor equips a specific pairwise comparison with inference. WAGER supplies image-cluster-robust confidence intervals and a cell-stratified randomization test for exactly the gain a benchmark submission claims. The technical contribution here is narrower than a new form of score decomposition: it is the transport construction and finite-sample identity that convert a known resolution concept into a model-to-model attribution algorithm with an exactness and inference guarantee that the single-model decomposition does not provide.

3. Method

3.1. Setup: audit a gain, not a model

Let $q_0(\cdot \mid x)$ and $q_1(\cdot \mid x)$ be the predictive distributions of an old and a new frozen model over K labels. Let $S(q, y)$ be a bounded proper score, oriented so that larger is better, and let $C = \phi(X)$ be a declared discrete prior feature. For SGG, C is the ordered subject–object class pair. Define the instance-specific *gain vector*

$$H_x(y) = S(q_1(\cdot \mid x), y) - S(q_0(\cdot \mid x), y), \qquad y = 1, \ldots, K. \tag{1}$$

Unlike the realized gain $H_X(Y)$, this vector can be evaluated at a counterfactual label without rerunning either model.

Our default is the quadratic score

$$S_{\mathrm{B}}(q, y) = 2q_y - \|q\|_2^2, \tag{2}$$

which differs from negative Brier loss only by a label-only constant. It is strictly proper and bounded, avoids probability clipping, and uses the entire predictive distribution. A floored log score is a sensitivity analysis.

3.2. Within-cell counterfactual transport

For a prior cell c, define three population quantities:

$$\Delta T_c = \mathbb{E}[H_X(Y) \mid C = c], \tag{3}$$

$$\Delta P_c = \mathbb{E}_{H|c}\, \mathbb{E}_{Y'|c}[H_X(Y')], \tag{4}$$

$$\Delta R_c = \Delta T_c - \Delta P_c. \tag{5}$$

Here Y' is an independent label drawn from the same cell. Thus ΔT_c is the observed score gain, ΔP_c is the gain retained after breaking the instance-specific prediction–label assignment while preserving the prior cell, and ΔR_c is the alignment gain lost by that transport.

Theorem 1 (Transport and covariance identities)**.** *For any integrable score contrast H,*

$$\Delta T_c = \Delta P_c + \Delta R_c, \qquad \Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c)\,. \tag{6}$$

For the quadratic score, writing $\Delta q_y(X) = q_1(y \mid X) - q_0(y \mid X)$,

$$\Delta R_c = 2 \sum_{y=1}^{K} \mathrm{Cov}\Big(\Delta q_y(X), \mathbb{1}\{Y = y\} \mid C = c\Big)\,. \tag{7}$$

Consequently, if the gain vector is a function of c alone, then $\Delta R_c = 0$.

Equation (7) is the operational meaning of WAGER: the new model receives alignment credit only to the extent that its probability change aligns more strongly with the label of the correct instance within the same prior cell. Cell-constant score improvements, including improved fitting of the cell's label histogram, remain in ΔP. The result does not assert that a nonzero covariance reflects causal or compositional understanding; unrecorded within-cell shortcuts can also contribute, so ϕ must be declared and stress-tested.

3.3. Generalization: the Bregman-score family

Nothing in the proof of Theorem 1 (Appendix Appendix A) uses properness: the identity $\Delta T_c = \Delta P_c + \Delta R_c$ and the covariance sum in Eq. (6) follow from the definitions of $\Delta T_c, \Delta P_c, \Delta R_c$ alone, for any score S. What is specific to the quadratic score is the further reduction to Eq. (7): the part of $H_x(y)$ that does not depend on y integrates to zero in the covariance sum because $\sum_y \mathbb{1}\{Y = y\} = 1$ almost surely. The same cancellation occurs for every strictly proper score generated by a Bregman divergence, so Eq. (7) and the log-score identity used as a robustness check in Section 4.5 are two named instances of one theorem rather than an unexplained extra case.

For a strictly convex, differentiable $\psi : \Delta_K \to \mathbb{R}$ on the probability simplex, the associated *Bregman score* is

$$S_\psi(q, y) = \psi(q) + \langle \nabla\psi(q),\, e_y - q \rangle, \tag{8}$$

where e_y is the y-th standard basis vector. For the true label distribution p,

$$\mathbb{E}_p[S_\psi(p, Y)] - \mathbb{E}_p[S_\psi(q, Y)] = \psi(p) - \psi(q) - \langle \nabla\psi(q), p - q \rangle = D_\psi(p, q) \geq 0, \tag{9}$$

the Bregman divergence of ψ between p and q, with equality iff $q = p$ by strict convexity, so S_ψ is strictly proper [19]. Taking $\psi(q) = \|q\|_2^2$ recovers the quadratic score of Eq. (2); taking $\psi(q) = \sum_k q_k \log q_k$ (negative entropy) recovers the log score.

Theorem 2 (Bregman-score covariance identity)**.** *Let $H_x(y) = S_\psi(q_1(\cdot \mid x), y) - S_\psi(q_0(\cdot \mid x), y)$ for a Bregman score generated by a strictly convex, differentiable ψ, and write $\Delta g_y(x) = \partial_y \psi(q_1(\cdot \mid x)) - \partial_y \psi(q_0(\cdot \mid x))$ for the contrast of the y-th gradient coordinate. Then*

$$\Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}(\Delta g_y(X),\, \mathbb{1}\{Y = y\} \mid C = c). \tag{10}$$

Corollary 1 (Quadratic and log score)**.** *For $\psi(q) = \|q\|_2^2$, $\Delta g_y(x) = 2\Delta q_y(x)$ and Eq.* (10) *is Eq.* (7)*. For $\psi(q) = \sum_k q_k \log q_k$, $\Delta g_y(x) = \log q_1(y \mid x) - \log q_0(y \mid x)$, and*

$$\Delta R_c = \sum_{y=1}^{K} \operatorname{Cov}\!\left(\log \frac{q_1(y \mid X)}{q_0(y \mid X)},\ \mathbb{1}\{Y = y\} \,\middle|\, C = c\right). \tag{11}$$

Eq. (11) gives the log-score check of Section 4.5 the same exact covariance meaning that Eq. (7) gives the quadratic score: instance alignment under the log score is the within-cell covariance between the log-likelihood-ratio contrast of the two models and the observed label indicator, not merely "a sensitivity analysis." Any other Bregman score – the spherical or power-family scores catalogued by Gneiting and Raftery [19], for instance – inherits an identity of the same form through its own ψ, so Theorem 2 fixes the scope of the estimator's scoring-rule dependence once, rather than leaving each alternative score to be checked separately.

3.4. Coarsening the audit cell

Section 4.4 reports that replacing the class-pair cell with a coarser subject-only cell changes the estimated alignment gain. The following result gives that sensitivity an exact population-level mechanism rather than treating it as an empirical curiosity.

Proposition 1 (Coarsening decomposition)**.** *Let $\bar{\phi} = g(\phi)$ be any coarsening of the audit feature, so that each cell $\bar{c}$ of $\bar{\phi}$ is a union of cells of ϕ. Then*

$$\Delta R_{\bar{c}} = \mathbb{E}[\Delta R_C \mid \bar{\phi} = \bar{c}] + \sum_{y=1}^{K} \operatorname{Cov}\big(\mathbb{E}[H_X(y) \mid C],\ \Pr(Y = y \mid C) \,\big|\, \bar{\phi} = \bar{c}\big)\,. \tag{12}$$

The first term is the coarse cell's share of the fine-grained alignment gain the paper already reports; it is the quantity a reader would expect a coarser audit to recover on average. The second term is new: it is a between-cell covariance, taken over the finer cells nested inside $\bar{c}$, between each fine cell's typical gain at label y and its label-y frequency. It is generally not sign definite, so coarsening ϕ can inflate or deflate the credited alignment gain relative to the finer audit, and Table 4 shows the empirically inflating case. The second term vanishes exactly when the quantities it correlates do not vary across the fine cells nested in $\bar{c}$ – that is, when $\mathbb{E}[H_X(y) \mid C]$ or $\Pr(Y = y \mid C)$ is constant within each coarse cell. Proposition 1 thereby formalizes why the finest defensible audit feature is the conservative default: relative to the finer audit, coarsening changes the estimate only by adding cross-cell prior structure into ΔR, while the finer partition's alignment gain is preserved on average through the first term. The proof is a per-label application of the law of total covariance, summed over y (Appendix Appendix A).

3.5. Sensitivity to an unrecorded confounder

Proposition 1 already describes exactly what happens when a relevant covariate is folded into ϕ: doing so refines the cell. Read in the other direction, the same identity quantifies what happens when a relevant covariate is *never* declared, which is WAGER's main scope limit (Section 5). Let Z be any unrecorded variable and let (ϕ, Z)

denote the finer, fully adjusted cell. Applying Proposition 1 with $\bar{\phi} = \phi$ and fine cell $C = (\phi, Z)$ gives, for every declared cell c of ϕ,

$$\Delta R_c(\phi) = \mathbb{E}\Big[\Delta R_{(\phi,Z)} \mid \phi = c\Big] + \sum_{y=1}^{K} \mathrm{Cov}(\mathbb{E}[H_X(y) \mid \phi, Z],\, \Pr(Y = y \mid \phi, Z) \mid \phi = c)\,. \tag{13}$$

The first term is what an analyst who could observe Z would report; the second is the bias of using ϕ alone. Eq. (13) is exact but requires knowing Z's joint law with (H, Y) to evaluate. The following bound turns it into a sensitivity analysis that requires only a judgment about how strongly an unrecorded variable could plausibly act, in the spirit of Rosenbaum [44] and recent partial-correlation sensitivity models for unmeasured confounding [16, 61].

Proposition 2 (Sensitivity bound for an unrecorded confounder)**.** *Suppose $H_x(y) \in [-M, M]$ for all x, y (WAGER already requires a bounded score). Fix a cell c of ϕ, write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$ (both conditional on $\phi = c$) for the two K-vectors on the right of Eq.* (13)*, and let*

$$\rho = \frac{\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)}{\sqrt{\sum_{y=1}^{K} \mathrm{Var}(a_y \mid c)}\sqrt{\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c)}} \in [-1, 1] \tag{14}$$

be their aggregate correlation. Then

$$\begin{aligned}\Big|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]\Big| \;&\le\; |\rho| \sqrt{\textstyle\sum_{y=1}^{K} \mathrm{Var}(a_y \mid c)}\; \sqrt{\textstyle\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c)} \\ &\le\; |\rho| \sqrt{\textstyle\sum_{y=1}^{K} \mathrm{Var}(H_X(y) \mid \phi = c)}\; \sqrt{\textstyle\sum_{y=1}^{K} p_y(c)(1 - p_y(c))},\end{aligned} \tag{15}$$

where $p_y(c) = \Pr(Y = y \mid \phi = c)$; the second inequality holds termwise ($\mathrm{Var}(a_y \mid c) \le \mathrm{Var}(H_X(y) \mid c)$ *and* $\mathrm{Var}(b_y \mid c) \le p_y(c)(1 - p_y(c))$ *by the law of total variance, since conditioning on more variables cannot increase the variance of a conditional expectation) followed by monotonicity of the square root applied to each sum separately – a product of two square-root sums, not a sum of per-label square roots, which the law of total variance alone would not license.*

Corollary 2 (Worst-case bound)**.** *Since* $\mathrm{Var}(a_y \mid c) \le M^2$ *and* $\mathrm{Var}(b_y \mid c) \le 1/4$ *by Popoviciu's inequality (each a_y ranges over $[-M, M]$ as a conditional expectation of H, each b_y over* $[0, 1]$*), Eq.* (15) *gives the data-free bound*

$$\Big|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]\Big| \;\le\; \frac{|\rho|\, K\, M}{2}. \tag{16}$$

An analyst willing to bound $|\rho| \le \bar{\rho}$ – how strongly an unrecorded variable could jointly move a model's within-cell score gain and the within-cell label distribution – obtains a numeric limit on how far $\widehat{\Delta R}_c$ can be from what a fully adjusted audit would find, without observing or modeling Z, in the spirit of Rosenbaum [44] and recent partial-correlation sensitivity models for unmeasured confounding [10, 16, 61]. Because $\bar{\rho}$ is

Table 1: Corollary 2's worst-case bound evaluated on the coarsening already reported in Table 4 ($K = 50$, $M = 2$ for the Brier score). The empirical bias is the law-of-total-expectation aggregate of Eq. (13)'s covariance term across all subject-only cells; `experiments/sensitivity_bound_check.py` reproduces it from the committed ablation results.

Quantity	Value
$\widehat{\Delta R}$, ϕ = subject-only	0.02181
$\widehat{\Delta R}$, ϕ = class-pair	0.01439
Empirical coarsening bias	0.00742
Crude bound, $KM/2$	50.00
Minimum $\bar{\rho}$ for the crude bound to hold	0.00015

a correlation rather than a domain-specific odds or risk ratio, it can be elicited the same way across benchmarks. Eq. (15)'s middle term uses the cell's own dispersion and is always at least as tight as Corollary 2's worst-case form, which needs only K and M and no per-cell computation at all; Table 1 evaluates the latter on VG150, where the coarsening from the class-pair cell to the subject-only cell instantiates Eq. (13) exactly, with Z = object class.

The crude bound holds for every $\bar{\rho} \geq 0.00015$, so it is directionally correct but, at VG150's $K = 50$, more than three orders of magnitude looser than the actual bias: any sensitivity parameter an analyst would plausibly declare (say $\bar{\rho} = 0.1$) certifies a limit far above what is ever observed. This is an honest property of the worst-case Popoviciu bound applied to K label probabilities that in practice concentrate well away from the $0/1$ extremes at which that inequality is tight, not a defect specific to this dataset. Proposition 2's second inequality is the fix: replacing the worst-case sums KM^2 and $K/4$ with the cell's own $\sum_y \mathrm{Var}(H_X(y) \mid c)$ and $\sum_y p_y(c)(1 - p_y(c))$ tightens the bound by exactly the ratio of the true aggregate dispersion to its worst case, without any further assumption on Z; `experiments/sensitivity_analysis.py` computes this sharper form from per-instance arrays, at the cost of needing them rather than only the committed cell-level aggregates the crude form reads directly. Even in its loose, data-free form, the bound gives a falsifiable prediction Proposition 1 alone does not: an unrecorded confounder would have to act with an implausibly strong, simultaneous effect on both channels to explain away a reported $\widehat{\Delta R}$ of the size seen in Section 4.3.

3.6. *Exact finite-sample decomposition*

Consider cell c with $n_c \geq 2$ examples. Write $H_i(y) = H_{x_i}(y)$. WAGER computes

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_{i \in c} H_i(y_i), \tag{17}$$

$$\widehat{\Delta P}_c = \frac{1}{n_c(n_c - 1)} \sum_{i \neq j \in c} H_i(y_j), \tag{18}$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j \in c} A_{ij}, \tag{19}$$

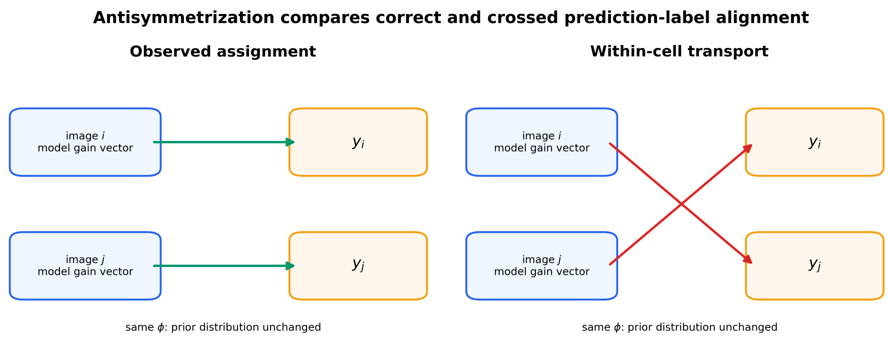

Figure 2: **The antisymmetric pair kernel.** For two examples in the same prior cell, WAGER subtracts the score under crossed labels from the score under the observed assignment. Cell membership and label counts are unchanged; only instance alignment is removed.

where the antisymmetric kernel is

$$A_{ij} = \frac{1}{2}\{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \tag{20}$$

Theorem 3 (Finite-sample WAGER identity)**.** *For every cell with $n_c \geq 2$, without a distributional assumption,*

$$\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c. \tag{21}$$

Moreover, when the cell's examples are drawn i.i.d. from the cell population, $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic for ΔR_c. Independence, not exchangeability alone, is what unbiasedness requires: the independent-coupling definition of ΔP_c pairs an example with a label drawn independently from the same cell, so $\mathbb{E}[H_1(Y_2)]$ matches it only for independent within-cell draws.

The dataset-level quantities weight cells by their number of identified examples: $\widehat{\Delta Z} = N_*^{-1} \sum_{c:n_c \geq 2} n_c \widehat{\Delta Z}_c$ for $Z \in \{T, P, R\}$. Singleton cells cannot identify a within-cell counterfactual; WAGER excludes and reports them instead of inventing a smoothed estimate. If $\widehat{\Delta T} > 0$, the descriptive alignment share is $\widehat{\rho} = \widehat{\Delta R}/\widehat{\Delta T}$. It may exceed one when instance alignment improves while prior fitting deteriorates, and it is undefined for a nonpositive total gain. We therefore report $\widehat{\Delta R}$ as the primary result, not a thresholded ratio.

3.7. Linear-time computation

Naively evaluating Eq. (18) is quadratic in cell size. Let m_{cy} be the count of label y in cell c. For example $i \in c$, its transported score is

$$P_i = \frac{\sum_{y=1}^{K} m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}. \tag{22}$$

Averaging P_i yields Eq. (18); averaging $H_i(y_i) - P_i$ yields Eq. (19). Thus the entire estimator costs $O(NK)$ time and $O(NK)$ storage for cached prediction vectors. No model fitting, projection fold, hierarchy, or smoothing hyperparameter appears.

3.8. Why leave-one-out transport, not a fitted-in-sample prior

Equation (22) excludes example i's own label from its transported score. It is natural to ask what is lost by the simpler alternative of fitting the cell's empirical label frequency on all n_c examples, including i, and using that fitted frequency as the counterfactual, i.e. treating $\hat{p}_c(y) = m_{cy}/n_c$ as a prior model and setting $\tilde{P}_i = \sum_y \hat{p}_c(y) H_i(y) = n_c^{-1} \sum_y m_{cy} H_i(y)$. This is the natural plug-in estimator, and it is what a frequency-collapsed projection baseline computes when no separate evaluation fold is withheld from fitting.

Proposition 3 (Exact attenuation of the in-sample plug-in)**.** *For every cell with $n_c \geq 2$ and every $i \in c$, writing $\hat{P}_i$ for Eq.* (22) *and $\hat{R}_i = H_i(y_i) - \hat{P}_i$,*

$$\tilde{P}_i = \hat{P}_i + \frac{1}{n_c}\hat{R}_i, \qquad \tilde{R}_i := H_i(y_i) - \tilde{P}_i = \frac{n_c - 1}{n_c}\,\hat{R}_i. \tag{23}$$

Consequently the in-sample plug-in's dataset-level statistic obeys $\widetilde{\Delta R} = N_*^{-1} \sum_{c:n_c \geq 2} (n_c - 1)\widehat{\Delta R}_c$, a multiplicative shrinkage of every cell's leave-one-out alignment gain by the factor $(n_c - 1)/n_c$.

The shrinkage in Eq. (23) is deterministic, not a variance statement: whenever $\widehat{\Delta R}_c \neq 0$, the in-sample plug-in is biased for it at every finite n_c, with the largest relative attenuation exactly in the smallest identified cells (50% at $n_c = 2$, vanishing only as $n_c \to \infty$). The standard remedy for this kind of self-prediction bias is a projection/evaluation split: a held-out fold restores unbiasedness by fitting $\hat{p}_c$ on data disjoint from the scored examples, but it does so by discarding examples from both stages and by introducing a split-fraction hyperparameter and, typically, higher variance. WAGER's leave-one-out U-statistic removes the bias exactly, at the full sample, for every ϕ-cell with $n_c \geq 2$, without a fitted nuisance distribution at all. Section 2.5 situates this result against the classical single-model score decomposition, where the analogous self-prediction bias affects each model's resolution estimate separately.

3.9. Inference

For point estimation we use all unordered within-cell pairs. For uncertainty, define $\bar{A}_i = (n_c - 1)^{-1} \sum_{j \neq i} A_{ij}$ and $\widehat{\Delta R}_c = n_c^{-1} \sum_i \bar{A}_i$. The estimated Hájek influence contribution for random cell membership is

$$\widehat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}. \tag{24}$$

We sum $\widehat{\psi}_i$ within image and use the resulting one-way sandwich variance, so multiple relations from the same image are not treated as independent. Total-gain influence is $H_i(y_i) - \widehat{\Delta T}$; prior-gain and share intervals follow from the exact identity and delta method.

As a complementary finite-group diagnostic, WAGER applies an independently sampled cyclic shift to labels inside every cell. Each shift preserves cell size and label histogram. Recomputing $\widehat{\Delta R}$ under B shifts gives

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{\widehat{\Delta R}^{(b)} \geq \widehat{\Delta R}\}}{B + 1}. \tag{25}$$

This is exact under the sharp null that labels are exchangeable relative to the frozen model outputs within each cell. Because SGG relations share images, our primary real-data uncertainty statement is the image-cluster-robust interval; the randomization test is reported as a secondary relation-level diagnostic.

3.10. Asymptotic normality

Eq. (24) and the sandwich variance used for the reported intervals are introduced in Appendix Appendix B through a Hájek-projection expansion that is stated as $o_p(N_*^{-1/2})$ but not, until now, given a limit theorem. The following result supplies one, under regularity conditions that hold whenever the score is bounded and clusters do not grow without bound relative to the sample.

Theorem 4 (Asymptotic normality of the alignment-gain estimator)**.** *Suppose (i) $H_x(y) \in [-M, M]$ for all x, y; (ii) examples are partitioned into clusters $g = 1, \dots, G$ (e.g. images) that are mutually independent, with cluster sizes having a finite third moment and $\max_g |g| = o(\sqrt{G})$; (iii) the identified fraction $N_*/N \to \kappa \in (0, 1]$ in probability; and (iv) the limiting variance $\sigma^2 = \lim_{G\to\infty} \mathrm{Var}(\sqrt{N_*}(\widehat{\Delta R} - \theta))$ exists and is strictly positive, where θ is the identified-subpopulation estimand of Appendix Appendix B. Then*

$$\sqrt{N_*}(\widehat{\Delta R} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \tag{26}$$

and the plug-in sandwich variance $\widehat{\sigma}^2 = (N_*/G) \sum_g (\sum_{i\in g} \widehat{\psi}_i/N_*)^2 \cdot G/(G-1)$ of Appendix Appendix B is consistent for σ^2.

The proof (Appendix Appendix A) combines the exact linearization of Appendix Appendix B with a Lindeberg–Feller central limit theorem for the independent-across-cluster sum of mean-zero, uniformly bounded terms [47, 53]; boundedness (i) makes the Lindeberg condition automatic given (ii), and consistency of $\widehat{\sigma}^2$ follows from the weak law of large numbers applied to each plug-in $\widehat{\Delta R}_c, \widehat{\Delta R}$ together with continuity of the variance functional. Condition (ii) is what rules out a single cluster dominating the sum – an image contributing a positive fraction of all relations, for instance – and is a standard requirement for cluster-robust inference generally, not specific to this estimator. This theorem is what licenses reading the image-clustered intervals reported throughout Section 4 as asymptotically valid rather than merely a plausible finite-sample proxy.

Relation to comparative forecast evaluation..

Corollary 3 (Relation to Diebold–Mariano/Giacomini–White)**.** *Under the trivial partition $\phi \equiv$ const (one cell containing every example), $\widehat{\Delta T} = N^{-1} \sum_i H_i(y_i)$ is the sample mean of the paired loss differential between the two models, and Theorem 4 applied to $\widehat{\Delta T}$ with its image-clustered interval is the grouped, cluster-robust form of the Diebold–Mariano statistic for that differential [13]; because the conditioning information set is empty, it is equivalently a degenerate instance of the Giacomini–White conditional predictive-ability test [18].*

Algorithm 1 WAGER gain decomposition

Require: frozen q_1, q_0; labels y; prior cells ϕ; proper score S
Ensure: $\widehat{\Delta T}, \widehat{\Delta P}, \widehat{\Delta R}$ and uncertainty

```
H_i(k) ← S(q_{1i}, k) − S(q_{0i}, k) for all i, k
for each cell c with n_c ≥ 2 do
    count labels m_{c1}, ..., m_{cK}
    for each i ∈ c do
        P_i ← (∑_k m_{ck} H_i(k) − H_i(y_i))/(n_c − 1)
        R_i ← H_i(y_i) − P_i
    end for
end for
ΔT^ ← mean H_i(y_i); ΔP^ ← mean P_i; ΔR^ ← mean R_i
compute image-clustered Hájek interval and cell-shift randomization p
```

This corollary is immediate from the definitions: DM and GW test whether a paired loss differential's (conditional) mean is zero, and $\widehat{\Delta T}$ at the trivial partition is exactly that differential's sample mean. $\widehat{\Delta P}$ and $\widehat{\Delta R}$ remain perfectly well defined at the trivial partition too — the single giant cell still supports leave-one-out label transport whenever $N \geq 2$ — but neither DM nor GW forms them: both tests target ΔT alone and stop there. $\widehat{\Delta R}$ at the trivial partition is then exactly the classical resolution term of Section 2.5's single-model decomposition, applied without any conditioning at all, and $\widehat{\Delta P}$ the corresponding calibration term, so WAGER's contribution at this special case is the further exact split of the DM/GW differential into the two components neither test provides. Conditioning on a nontrivial ϕ is what separates WAGER's estimand from the classical target more generally: DM/GW ask whether one forecaster beats another on average, optionally conditional on a test function of an information set [18]; WAGER asks the same question at every declared cell *and* attributes the answer, cell by cell, into a component recoverable from the label histogram alone and a component that is not. Every real-data comparison in Section 4 could equivalently be read as reporting a clustered DM statistic for $\widehat{\Delta T}$ (Corollary 3) alongside the two-way split that DM/GW do not provide.

4. Experiments

We evaluate five questions: (i) does WAGER remain centered under the null and respond monotonically to known instance signal; (ii) does it distinguish class-prior fitting from spatial evidence on a real long-tailed benchmark; (iii) are conclusions stable to score, cell granularity, and sparse-cell filtering; (iv) does the alignment channel behave sensibly when a model consumes real image pixels rather than annotation-derived geometry; and (v) does the same estimator, unchanged, produce a meaningful attribution on a structurally different task in another domain? Questions (i)–(iii) use Visual Genome and controlled simulations; (iv) adds a frozen CLIP encoder on real VG images; (v) moves to long-tailed image classification on CIFAR-100-LT. All confidence intervals are 95%.

4.1. Controlled validation

We generate 24 prior cells over five labels, drawing each cell's label distribution from a Dirichlet law and taking the old model to be exactly the cell prior. The new model interpolates toward an oracle, $q_1 = (1-\beta)q_0 + \beta q_{\text{oracle}}$ with q_{oracle} placing 0.98 on the realized label, and forty repetitions are run at each $\beta \in \{0, .1, \dots, .8\}$. Across that range the estimated alignment gain grows linearly and monotonically with the injected signal (Figure 3, left).

Type-I behaviour is assessed by perturbing the prior predictions with mean-zero noise and then randomly reassigning the perturbed vectors within each cell, which guarantees that no systematic prediction–label alignment survives. Over 200 such datasets the 99-shift randomization test rejects in 0.030 of runs at level 0.05, with a mean estimated alignment gain of -8.97×10^{-5}; the null p-values come out approximately uniform and the null estimates stay centred at zero (Figure 3, centre and right).

Coverage of the primary interval.. The oracle-signal and null designs above validate the estimator's centering and the randomization diagnostic, but the paper's primary uncertainty statement is the image-clustered Hájek interval, so we validate its coverage directly in the regime that stresses it: relations grouped into images (one to about thirty per image) whose oracle weight shares a latent per-image effect, and Zipf-distributed cell sizes in which most identified cells hold exactly two examples. Against the design expectation estimated from 4,000 independent replications, the cluster-robust interval covers in 94.0% of 500 runs, while an interval that wrongly treats relations as independent covers in only 88.6% – quantifying both the mild finite-sample undercoverage of the asymptotic interval and the necessity of clustering.

Discriminant validity.. Three further designs test that the channels separate what they claim to separate. First, a *calibration-only* alternative: the new model is a temperature-softened copy of an overconfident informative model, so the pair contains identical instance information. The raw decomposition credits it with a large negative alignment gain (mean −0.488) – the confound analyzed in Section 4.7 – while the calibration-matched protocol used there reduces it to −0.0003 (sd 0.0017 over 200 runs), confirming that the protocol removes exactly the confidence-scaling channel. Second, a *prior-only* improvement (the new model corrects a heteroscedastically noisy estimate of the cell prior, adding no instance information) yields an alignment gain centred at zero (0.0002, sd 0.0035) with a clearly positive prior channel: prior improvements do not leak into $\widehat{\Delta R}$ beyond the algebraic guarantee of Theorem 1. Third, an *unrecorded within-cell shortcut* (a binary feature, varying inside each cell, that shifts the label distribution and is used by the new model) is credited to alignment (+0.041, sd 0.004), quantifying the stated limitation that $\widehat{\Delta R}$ measures all within-cell predictive signal relative to the declared ϕ, shortcut or otherwise, and cannot by itself distinguish the two.

4.2. Visual Genome setup

We work with Visual Genome [30] in the PredCls regime [56], where ground-truth boxes and object labels are supplied and the model has only to predict one of 50 predicate labels for each ordered object pair. The predictors of this section are audited on a

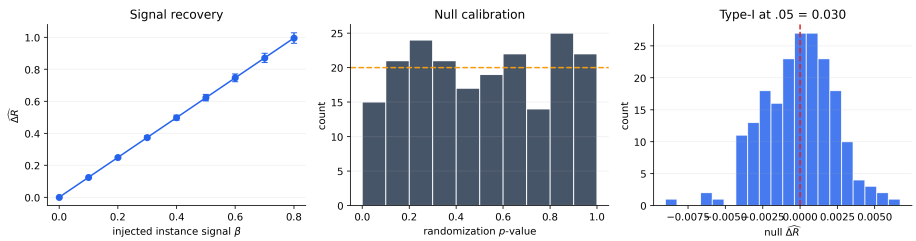

Figure 3: **Controlled validation.** WAGER recovers injected within-cell signal monotonically (mean ± one standard deviation), produces approximately uniform null randomization p-values, and controls type-I error in 200 null runs.

VG150-*style* dataset we build directly from the released relationship annotations: the 150 most frequent object classes and 50 most frequent predicates form the vocabulary, and images are split 70/30 into a pool for fitting the compared predictors and a pool on which the audit runs. We use this reconstruction rather than the canonical VG150 release because these predictors are trained here from annotations alone; Section 4.5 audits *released third-party checkpoints* on the canonical split itself, so both settings appear in the paper and neither is asked to stand for the other. The reconstruction provides 532,987 training and 229,605 test relations, drawn from 65,228 and 27,955 images respectively, and the prior cell throughout is ϕ = (subject class, object class), two real examples of which appear in Figure 4. Of the 8,614 cells present in the test split, 6,346 contain at least two relations and are therefore identified; the remaining 2,268 are singletons, which WAGER excludes and reports rather than smoothing over, leaving 227,337 relations (99.0%) under audit.

The five predictors are deliberately constructed so that each exposes a different kind of information. FREQ is the training-count distribution conditioned on the class pair, with add-one smoothing and a subject-marginal and then global backoff for class pairs unseen in training; FREQ+OVERLAP adds to it a single binary box-overlap indicator. MLP-CLASS consumes fixed subject and object class embeddings, which makes it a function of ϕ alone; MLP-SPATIAL extends those embeddings with 12 normalized box-geometry features covering relative centre, scale, aspect, intersection, containment and IoU, while MLP-SPATIAL+ both widens the network and trains longer (Appendix Appendix D lists every architecture and schedule), so its increment over MLP-SPATIAL reflects capacity and optimization jointly rather than capacity alone. Each is trained once on the training split, and WAGER thereafter consumes nothing but their cached test distributions, scored quadratically with image-clustered intervals and 499 within-cell cyclic randomizations, whose smallest attainable p-value is $1/500 = .002$; table entries of .002 are at this resolution floor and should be read as $p \leq .002$. With confidence intervals as the primary uncertainty statement and every significant effect many standard errors from zero, we do not additionally adjust the diagnostic p-values for the number of reported comparisons.

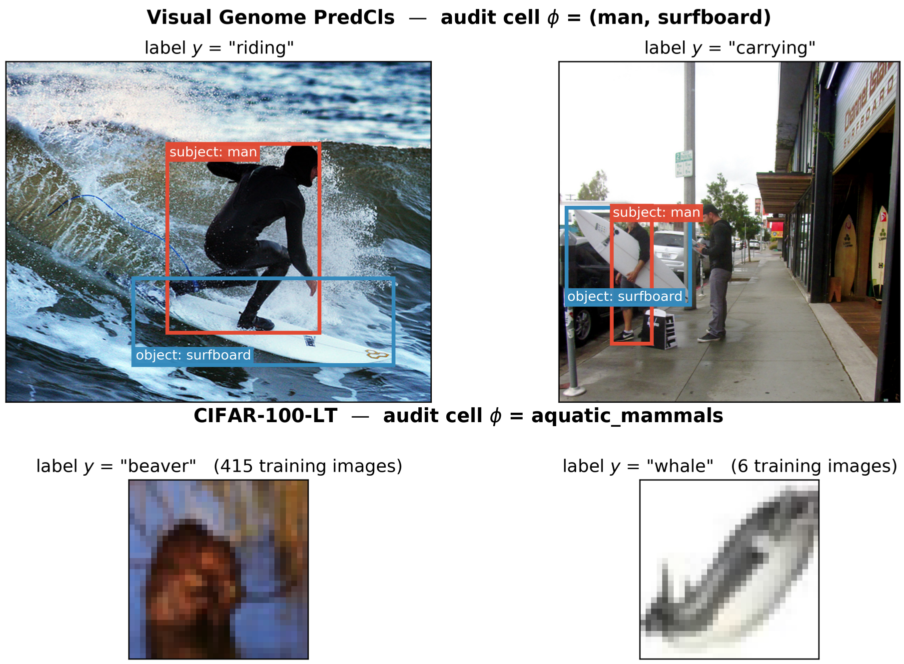

Figure 4: **What an audit cell contains, on both benchmarks.** Each row shows two real test examples that share the same declared prior feature ϕ but carry different labels – precisely the configuration within-cell transport exploits. *Top:* two Visual Genome relations from the cell ϕ = (man, surfboard). The class pair is identical and therefore cannot distinguish "riding" from "carrying"; only the image content can, which is what ΔR is designed to credit. Boxes are the annotated subject (red) and object (blue) regions that MLP-SPATIAL turns into geometry features and MLP-VISUAL crops and encodes with CLIP. *Bottom:* two CIFAR-100 test images from the superclass ϕ = aquatic_mammals, drawn from opposite ends of the long-tailed training distribution (415 versus 6 training images). Crossing the labels within a row preserves the cell and its label histogram while destroying which prediction belongs with which instance.

4.3. Direct gain attribution

The central result is collected in Table 3 and Figure 5. MLP-CLASS improves on FREQ by 0.03386 of quadratic score, yet because its gain vector is constant inside every class-pair cell, the estimator assigns that entire improvement to prior transport and precisely nothing to instance alignment. The zero is worth dwelling on: it follows algebraically from the construction rather than from a learned baseline happening to pass a sanity check.

Adding box geometry changes the picture. MLP-SPATIAL gains 0.04270 over FREQ, of which 0.02832 survives label transport and 0.01439 is instance alignment (CI $[0.01373, 0.01504]$). More revealing still is the direct comparison against MLP-CLASS, where the total gain falls to 0.00885 only because prior fitting deteriorates by 0.00554 even as spatial alignment improves by 0.01439. The resulting alignment share of 1.63 records a genuinely useful instance-specific change that the aggregate score partly conceals behind a worse prior component, and it is exactly the configuration that a ratio confined to $[0, 1]$, or thresholded at 0.5, would misdescribe.

Table 2: Frozen-model top-1 accuracy on VG150 PredCls. Similar accuracies do not reveal whether the difference is prior-recoverable.

Model	Accuracy
FREQ	0.6564
FREQ+OVERLAP	0.6545
MLP-CLASS	0.6554
MLP-SPATIAL	0.6639
MLP-SPATIAL+	0.6674

Table 3: WAGER decomposition on 227,337 identified VG150 PredCls relations. Gains use the quadratic score. CI is image-cluster robust; p is the 499-shift relation-level randomization diagnostic (resolution floor .002). "–" denotes an undefined share because total gain is negative. The MLP-CLASS row's zero-width interval is exact by construction, not a sampling statement: a gain vector constant within every cell makes the antisymmetric kernel identically zero (Theorem 1), so no variability exists for the interval to measure.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	$\widehat{\rho}$	p
FREQ+OVERLAP	FREQ	−0.01229	−0.01430	0.00202 [0.00168, 0.00235]	–	.002
MLP-CLASS	FREQ	0.03386	0.03386	0.00000 [0.00000, 0.00000]	0.00	.724
MLP-SPATIAL	FREQ	0.04270	0.02832	0.01439 [0.01373, 0.01504]	0.34	.002
MLP-SPATIAL+	FREQ	0.04644	0.02885	0.01760 [0.01686, 0.01834]	0.38	.002
MLP-SPATIAL	MLP-CLASS	0.00885	−0.00554	0.01439 [0.01373, 0.01504]	1.63	.002
MLP-SPATIAL+	MLP-SPATIAL	0.00374	0.00053	0.00321 [0.00289, 0.00353]	0.86	.002

The remaining two comparisons bracket the interesting range. MLP-SPATIAL+ adds 0.00374 over MLP-SPATIAL, nearly all of it alignment at 0.00321 (CI $[0.00289, 0.00353]$), whereas FREQ+OVERLAP supplies the opposite caution: its overlap indicator does create a small positive alignment term, but the accompanying negative prior term is larger, so the model ends up worse overall. Using image-derived information is thus never treated as an improvement in itself; both channels are reported, signs included.

Reading the magnitudes.. Quadratic-score units are not ones most readers calibrate by eye, so it is worth anchoring them in the rank-based metrics this literature reports. Where both channels point the same way the correspondence is straightforward: MLP-SPATIAL-S's $\widehat{\Delta R} = +0.01023$ over MLP-CLASS-S accompanies +0.51 points of top-1 predicate accuracy, +0.60 of mean reciprocal rank and +0.86 of recall@5, so in these data a hundredth of quadratic-score gain is worth roughly half an accuracy point. The alignment channel is not, however, a re-description of any rank metric, and the real-pixel comparison of Section 4.6 is where that becomes visible: there $\widehat{\Delta R} = +0.00641$ while top-1 accuracy falls by 3.07 points and recall@5 by 2.21. Prior-matching both models before scoring narrows the accuracy gap to 2.17 points without closing it. The rank metrics aggregate the two channels into one number; when the channels carry opposite signs, as they do here, no single such number can represent both, which is precisely the situation the decomposition exists to report.

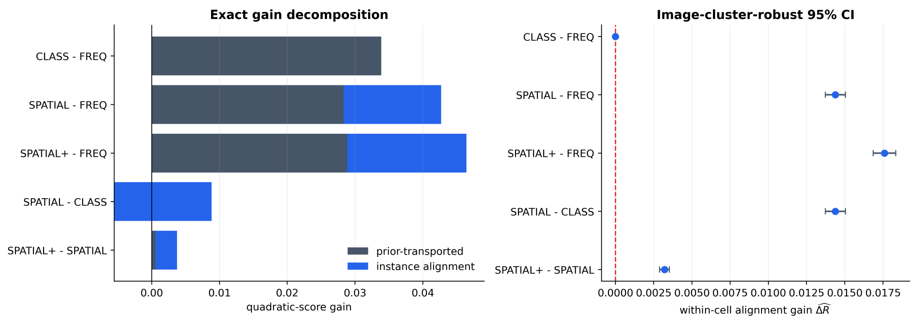

Figure 5: **Direct model-pair attribution.** Left: exact decomposition of each positive gain into prior-transported and instance-alignment components (the negative prior term for SPATIAL–CLASS extends left of zero). Right: primary alignment estimates with image-cluster-robust intervals.

Reading $\widehat{\Delta T}$ as a comparative forecast test.. Corollary 3 identifies $\widehat{\Delta T}$ and its image-clustered interval in Table 3 as exactly a grouped Diebold–Mariano statistic and interval for each pair's paired score differential, so every row already reports what that classical test would report. What it does not report is the next two columns: whether MOTIFS-TDE (Section 4.5) or MLP-SPATIAL beat their baseline for a reason a frequency-only remodeling of the same benchmark could not reproduce, which is the question $\widehat{\Delta P}$ and $\widehat{\Delta R}$ answer and DM/GW do not ask.

4.4. Sensitivity analyses

For MLP-SPATIAL versus MLP-CLASS, the conclusion survives every planned sensitivity check (Table 4). Changing from the quadratic to a probability-floored log score changes scale (0.01439 to 0.06536) but not sign or significance. Coarsening ϕ from the class pair to subject class increases the credited alignment to 0.02181. Proposition 1 identifies why: the coarser cell's estimate equals the class-pair estimate's conditional average plus a between-class-pair covariance term, and that term is positive here, meaning some of the credited gain reflects class-pair-level structure that the finer audit had already correctly assigned to ΔP. This is precisely the sense in which the class-pair audit is the conservative choice: it is not that coarsening is wrong, but that it cannot be assumed neutral, and only the finer partition's estimate isolates alignment gain net of every prior regularity expressible in ϕ. The same class-pair-versus-subject-only pair also instantiates Proposition 2 directly, with the subject class playing the role of the unrecorded confounder Z: Table 1 in Section 3.5 reports that the resulting bias of 0.00742 is comfortably inside the Cauchy–Schwarz bound, though the bound is loose enough at $K = 50$ that it would remain satisfied for a $\bar{\rho}$ three orders of magnitude smaller than one. Raising the minimum cell size from 2 to 5, 10, and 20 yields 0.01395, 0.01330, and 0.01225; all cluster-robust lower limits remain positive. The procedure itself has no smoothing strength, split fraction, or decision threshold to tune.

4.5. Auditing a published debiasing method

The predictors audited so far were built to expose particular kinds of information. The question the method exists to answer, however, concerns systems that other peo-

Table 4: Sensitivity of the MLP-SPATIAL vs. MLP-CLASS decomposition to score, audit granularity, and minimum cell size. CI is image-cluster robust on $\widehat{\Delta R}$.

Factor	Setting	$\widehat{\Delta T}$	$\widehat{\Delta R}$ (95% CI)	Cells
Score	Brier (default)	0.00885	0.01439 [0.01373, 0.01504]	6346
Score	Log (floored)	0.04297	0.06536 [0.06294, 0.06779]	6346
Audit cell	class pair (default)	0.00885	0.01439 [0.01373, 0.01504]	6346
Audit cell	subject only	0.00943	0.02181 [0.02079, 0.02283]	150
Min. cell size	2 (default)	0.00885	0.01439 [0.01373, 0.01504]	6346
Min. cell size	5	0.00772	0.01395 [0.01329, 0.01462]	4060
Min. cell size	10	0.00683	0.01330 [0.01262, 0.01398]	2744
Min. cell size	20	0.00570	0.01225 [0.01155, 0.01295]	1757

ple release and that the field already compares, so we now apply WAGER unchanged to a published pair. Tang et al. [49] release a causal MOTIFS PredCls checkpoint that yields two models from one set of weights: the biased baseline (MOTIFS-SUM) and its Total Direct Effect counterpart (MOTIFS-TDE), which subtracts a context-only counterfactual at inference. TDE is among the most cited debiasing methods in scene graph generation, and it is reported through mean recall, so it is exactly the kind of claimed improvement whose composition is at issue.

We evaluate both variants with the authors' own code on the canonical VG150 test split, which fixes the vocabulary, the split and the relation set, and makes the two prediction sets aligned relation-for-relation by construction. Our reproduction matches the published behaviour: the baseline reaches $R@50 = 0.6612$ with $mR@50 = 0.1459$, and TDE trades the former for the latter, reaching $R@50 = 0.4588$ with $mR@50 = 0.2476$ and raising zero-shot recall from $zR@50 = 0.1102$ to 0.1432. Top-1 predicate accuracy falls from 0.6981 to 0.5577. The audit uses 183,639 relations over 26,446 images and 7,127 class-pair cells, of which 98.9% are identified.

Result.. Table 5 decomposes the pair. Read raw, TDE loses 0.11651 of quadratic score, of which 0.10296 sits in the prior channel and a small but significant 0.01355 in instance alignment. That reading does not survive the control this paper insists on. The two models differ sharply in confidence — the baseline is overconfident at $T^* = 2.50$ while TDE, having subtracted a counterfactual, is already near-calibrated at $T^* = 0.92$ — and once temperatures are fitted on a held-out half of the images and the audit runs on the other half, the alignment channel is indistinguishable from zero: $\widehat{\Delta R} = -0.00006$ with a 95% interval of $[-0.00120, +0.00109]$ and a randomization p of .555. Essentially the entire proper-score difference, -0.18258 of -0.18264, is prior-transported.

This is a precise and, we think, fair characterization of what TDE does. Its own construction subtracts the context-only prediction, so a method that operates on the prior channel is what its authors built; WAGER quantifies that the intervention is almost purely of that kind, and that the model's ability to tell which relation belongs to which image within a class-pair cell is left essentially untouched. The consequence for evaluation is the paper's thesis in a published setting: an improvement of 10.2 points of mean recall, a metric the field reads as evidence of better handling of rare predicates, is

Table 5: WAGER audit of released checkpoints on the canonical VG150 PredCls split: MOTIFS-TDE against the MOTIFS-SUM baseline it debiases, both evaluated with the original authors' code. Raw rows use all identified relations; the calibration-matched rows fit each model's temperature on a held-out half of the images and audit the disjoint half. CI is image-cluster robust on $\widehat{\Delta R}$.

Regime	Score	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)
raw	quadratic	class pair	−0.11651	−0.10296	−0.01355 [−0.01504, −0.01206]
raw	log	class pair	+0.03277	+0.15897	−0.12619
raw	quadratic	subject	−0.11536	+0.10175	−0.21711
calibration-matched	quadratic	class pair	−0.18264	−0.18258	−0.00006 [−0.00120, +0.00109]
calibration-matched	log	class pair	−0.54760	−0.59157	+0.04396

compatible with no measurable change in within-cell instance alignment. Mean recall and the alignment channel are answering different questions, and only the latter is a statement about instance-level evidence.

Two honest qualifications. First, the alignment conclusion is stable in sign under the audit cell but not in magnitude under the score: the calibration-matched log-score decomposition puts alignment at +0.04396 against a prior channel of −0.59157, so the claim we make is the one both scores support — that the prior channel carries the overwhelming majority of the difference — rather than a precise value for the remainder. Second, TDE is designed to change the decision rule rather than to emit calibrated probabilities, and a proper score evaluates the latter; the calibration-matched regime is what makes the comparison meaningful at all, and its necessity here is the clearest practical argument for the protocol of Section 4.7.

4.6. Real-pixel signal on Visual Genome

None of the models in Sections 4.2–4.4 ever sees an image. They are functions of the declared class pair and, in the case of MLP-SPATIAL(+), of box coordinates recorded in the annotation. Since box geometry serves only as a proxy for what an image contains, a sceptical reader is entitled to ask whether the alignment channel is responding to genuine pixel evidence or merely to that proxy. Figure 4 sharpens the question: the cell $\phi = (\text{man}, \text{surfboard})$ holds both "riding" and "carrying", and although their box configurations certainly differ, what actually distinguishes the two relations is what the photograph shows.

MLP-VISUAL is introduced to answer it. Subject and object regions are cropped from the original Visual Genome image and each is encoded by a frozen CLIP ViT-B/32 image encoder [43], after which the two 512-dimensional embeddings are concatenated with the class embeddings that MLP-CLASS already uses and passed to a small trained head of 256 and 128 units. Because CLIP itself is never fine-tuned, whatever alignment the model gains reflects information its pretraining had already recovered from pixels and the head has merely matched to an SGG label, rather than representation learning performed on Visual Genome.

Matched training data.. We report this comparison apart from Table 3 rather than as an additional row, for a reason that is itself an instance of the paper's argument. Encoding every training relation with CLIP is expensive enough that MLP-VISUAL must be trained on a subsample, and setting a subsampled model against the full-data MLP-SPATIAL would confound feature content with training-set size so thoroughly that a disappointing result could not honestly be attributed to the pixels. All three compared predictors are therefore retrained on a single fixed, seeded subsample of 100,000 of the 532,987 training relations, and are written MLP-CLASS-S, MLP-SPATIAL-S and MLP-VISUAL-S to mark the fact. Differing only in their feature sets, any gap between them isolates the features. The audit continues to use the complete 229,605-relation test split with the same class-pair ϕ and image-clustered intervals, so its numbers remain directly comparable with the main analysis. The sampling procedure, the crop-deduplication scheme and the count of images that could not be retrieved are given in Appendix Appendix D.

Result.. Judged by any aggregate measure, the CLIP model is the worst of the three. Its top-1 accuracy is 0.6161 against 0.6467 for MLP-SPATIAL-S and 0.6416 for MLP-CLASS-S, and its quadratic score is lower still, falling 0.04655 below MLP-SPATIAL-S. An evaluator working from either number would conclude that replacing box geometry with pixel content had simply made the predictor worse, and would discard it.

The decomposition in Table 6 and Figure 6 contradicts that reading on the axis the paper cares about. Against MLP-SPATIAL-S the CLIP model loses 0.05296 of prior-transported gain while *gaining* 0.00641 of instance alignment (95% CI $[0.00514, 0.00769]$, randomization $p = .002$), so its entire deficit sits in the prior channel and its instance-level discrimination is significantly better, not worse. The comparison against the class-only baseline tells the same story more plainly: pixels buy 0.01665 of alignment where box geometry buys 0.01023, and the two intervals are far apart. Real image content therefore carries more instance-specific signal than the geometric proxy standing in for it, which is precisely what Section 4.6 set out to test.

Why the prior channel collapses has a plausible mechanism, though we state it as a hypothesis rather than a finding. The 1024 CLIP dimensions dominate the 64 dimensions of class embedding that the head also receives, so the model fits the class-pair frequency structure of VG150 markedly less well than a predictor built around that structure. A benchmark whose headline metric rewards prior fitting will punish this model, and the aggregate numbers duly do. What WAGER adds is the observation that the punishment is not for failing to use the image; it is for failing to reproduce the annotation prior, while using the image better than either competitor.

The deficit and the advantage respond to different interventions.. If the two channels measure what they claim, they should respond asymmetrically to a post-hoc correction that supplies prior information but no instance information. We test this by prior matching: within each class-pair cell, a model's predictions are reweighted so that their cell mean matches the training-count histogram, a label-shift-style correction that uses no test labels. The correction registers, as it must, almost entirely in the prior channel ($\Delta P = +0.02401$ against $\Delta R = +0.00111$ for the CLIP model relative to its uncorrected self), and it cuts the CLIP model's aggregate deficit roughly in half (-0.04655 to -0.02143) while leaving the alignment advantage intact (+0.00752, or +0.00663 with both models corrected). A calibration-matched audit – temperatures fitted on a

Table 6: Matched-subsample real-pixel study on the full VG150 test split (coverage 99.0%, as in Table 3). All three predictors are trained on the same 100,000 relations and differ only in features. CI is image-cluster robust on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic.

New model	Old model	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
MLP-SPATIAL-S	MLP-CLASS-S	0.00503	−0.00520	0.01023 [0.00967, 0.01079]	.002
MLP-VISUAL-S	MLP-CLASS-S	−0.04152	−0.05816	0.01665 [0.01534, 0.01796]	.002
MLP-VISUAL-S	MLP-SPATIAL-S	−0.04655	−0.05296	0.00641 [0.00514, 0.00769]	.002

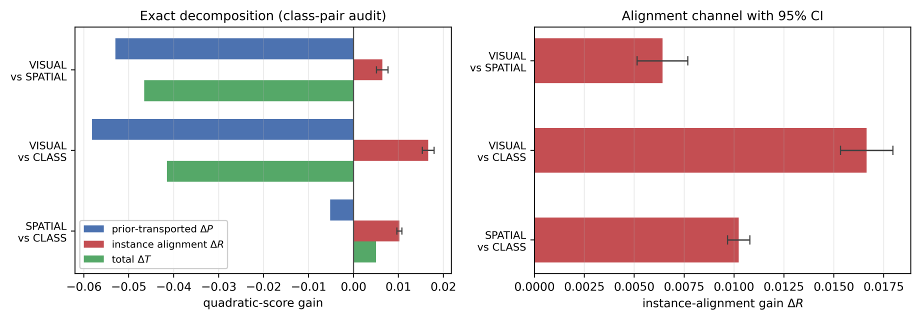

Figure 6: **Pixels beat geometry on alignment while losing on the aggregate.** Left: the exact decomposition for the three matched-subsample pairs. Both comparisons involving the CLIP model have strongly negative totals (green) driven entirely by the prior channel (blue), while their alignment channel (red) is positive in every case. Right: the alignment channel alone, with 95% image-clustered intervals. Replacing box geometry with real image content raises instance alignment from 0.01023 to 0.01665 over the same class-only baseline.

held-out image split, $T^* = 1.18$ against 1.00, so these heads are nearly calibrated to begin with – likewise leaves the advantage significant at +0.00467 [+0.00314, +0.00621]. The alignment channel is therefore neither manufacturable by prior correction nor an artifact of confidence scaling. Conversely, the part of the deficit that prior matching removes is exactly the part a deployment could restore by re-weighting; the remainder reflects differences in within-cell prediction dispersion rather than in the fitted cell histogram.

This is the paper's argument running in the opposite direction from Section 4.7. There a near-zero total gain concealed a large alignment *loss*; here a clearly negative total gain conceals a real alignment *gain*. In both cases the aggregate score is not merely imprecise but actively misleading about the thing a vision benchmark is meant to measure, and in both cases the two channels recover what it hides.

4.7. Cross-domain validation: long-tailed image classification

Sections 4.2–4.6 are all instances of one task (predicate classification on an ordered object pair). Section 2.2 argued that long-tailed image classification poses a structurally identical question: a class-frequency prior can be fit better without any per-instance improvement. We test this directly on CIFAR-100-LT [11], an exponential-imbalance (ratio 100) subsample of CIFAR-100's training split with the original, balanced test split retained for evaluation. We train ResNet-32 [22] classifiers that differ only in their per-class loss weighting: a vanilla cross-entropy baseline (CE); class-balanced effective-number re-weighting applied from initialization (CB) [11]; and the same re-weighting deferred to the final twenty epochs (DRW) [7]. Optimizer, schedule, augmentation, epoch budget, and seed are identical across runs (Appendix Appendix D).

Choosing ϕ.. A declared prior feature has to predict the label without determining it, and this second benchmark makes the constraint unusually easy to violate. Were ϕ taken to be the fine class label itself, every audit cell would be label-homogeneous, within-cell transport would have nothing to cross, and $\widehat{\Delta R}$ would come out identically zero – an algebraic artifact wearing the appearance of a finding. CIFAR-100's own 20 coarse superclasses avoid this, each gathering five fine labels into a cell of 500 balanced test images, which is structurally the situation of a VG class-pair cell holding several predicates (Figure 4). Alongside the superclass audit we report two further declared priors: the trivial coarsening to a single global cell, which audits against the global class marginal and so instantiates Proposition 1 in a second domain, and a partition into head, medium and tail training-frequency tiers. CIFAR-100 test images are independent, unlike the several relations VG draws from one photograph, so no cluster grouping is required and the plain Hájek interval of Section 3.9 applies directly. One semantic point matters in this domain: transport uses each cell's *test* label histogram, which on CIFAR-100-LT is balanced. A positive ΔP here therefore records movement *toward* the balanced test marginal – undoing the training-frequency skew – which is precisely the adjustment long-tailed evaluation is designed to reward; a prior-transported gain is a description of where score was earned, not an accusation (Section 5).

Result.. The three classifiers reach top-1 accuracies of 0.3662 (CE), 0.2627 (CB) and 0.3874 (DRW), so re-weighting from initialization degrades the classifier substantially while the deferred schedule delivers the improvement it was designed to produce. Table 9 decomposes both gains against the superclass audit.

Of the two, the class-balanced run is the more instructive, precisely because its aggregate score conceals almost everything of interest. On the quadratic score it gains +0.00143 over the baseline, a difference small enough that a reader working from the headline number alone would reasonably conclude that re-weighting had changed very little. The decomposition tells another story: within superclasses the prior-transported channel improves by +0.19713 while instance alignment deteriorates by -0.19569 (95% CI $[-0.20728, -0.18411]$), so the near-zero total is not a small effect but two large ones that happen to cancel.

A calibration-matched regime.. A channel split of that shape could, however, be produced by confidence scaling alone. Under the quadratic score the alignment channel is a covariance between probability movements and labels (Eq. (7)), and a monotone recalibration moves it: softening an overconfident model transfers score mass from the alignment to the prior channel while adding no information about any individual image. The baseline invites exactly this concern, assigning a mean maximum probability of 0.764 while being correct on only 0.3662 of the test set. We therefore repeat the audit in a calibration-matched regime: each model's temperature is chosen by held-out

Table 7: Raw and calibration-matched decompositions on CIFAR-100-LT at the superclass audit. Temperatures are fitted by likelihood on a held-out calibration half of the test split; every decomposition uses the disjoint audit half; entries are means (sd) over twenty random splits. The final row is the control: recalibrating the baseline alone carries no instance information yet moves both channels substantially.

Comparison	Regime	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$
CB vs CE	raw	+0.00208 (0.00642)	+0.19635 (0.00369)	−0.19427 (0.00632)
CB vs CE	calibration-matched	−0.12912 (0.00373)	+0.11079 (0.00372)	−0.23991 (0.00491)
DRW vs CE	raw	+0.06712 (0.00865)	+0.04258 (0.00463)	+0.02454 (0.00685)
DRW vs CE	calibration-matched	+0.02173 (0.00275)	+0.00216 (0.00410)	+0.01957 (0.00329)
CE recalibrated vs CE	control	+0.20495 (0.00420)	+0.37960 (0.00268)	−0.17464 (0.00455)

likelihood on a calibration half of the test split ($T^* = 2.85$ for CE, 3.51 for CB, 2.61 for DRW), the decomposition is computed on the disjoint audit half, and the whole procedure is repeated over twenty random splits (Table 7). The regime's necessity is quantified by its control row: recalibrating the baseline alone – a transformation carrying no instance information – moves the channels by +0.380 (prior) and −0.175 (alignment), so the raw split does mix calibration with discrimination.

The calibration-matched rows then separate the two schedules cleanly, and more sharply than the raw ones. CB's alignment deficit does not shrink but widens, to −0.23991 (0.00491): the model re-weighted from initialization has genuinely lost within-superclass discrimination – a real degradation that the ten-point accuracy drop records independently – and its near-zero raw total reflects that loss offset by its softer, better-calibrated confidence. DRW inverts the reading its raw split suggests. Raw, its +0.06712 gain divides roughly two-thirds prior to one-third alignment; calibration-matched, the improvement is +0.02173 (0.00275) of which +0.01957 (0.00329) is instance alignment while the prior channel is close to zero (+0.00216 (0.00410)). What survives calibration matching is precisely the component the deferred schedule was designed to buy – per-instance discrimination, concentrated on rare classes (Table 10) – while the apparently prior-recoverable share of its raw gain was largely the calibration difference between the two runs. Neither an accuracy difference nor an aggregate score exposes either composition, which is the practical argument for reporting the two channels, in both regimes, instead of their sum.

Seeds, imbalance ratios, and a post-hoc arm.. A decomposition computed from one trained pair carries only test-set sampling uncertainty, which is not the uncertainty relevant to a claim about a training *method*. We therefore retrained the triple at two further seeds at the primary ratio and at imbalance ratios 10 and 50, and added a fourth arm: post-hoc logit adjustment (LA), which divides the baseline's predictions by the training class prior without retraining [40]. Every configuration is audited in both regimes; Table 8 reports the calibration-matched alignment channel, which is the one the preceding paragraphs interpret.

Three things follow. First, the class-balanced run's alignment loss is not a misconfiguration but a systematic function of imbalance severity: −0.069 at ratio 10, −0.244 at 50, and −0.251 (0.011) across three seeds at 100, tracking an accuracy penalty that grows from 2.3 to 10.4 points over the same range. Second, the deferred schedule's alignment gain is positive at every ratio and in every seed, but its magnitude varies

Table 8: Calibration-matched alignment gain across seeds and imbalance ratios, with balanced test accuracy for each arm. LA is post-hoc logit adjustment applied to the baseline. The class-balanced deficit scales with imbalance severity; the deferred schedule's gain is positive throughout but seed-sensitive in magnitude; the post-hoc correction lands in the alignment channel rather than the prior channel.

		Top-1 accuracy				$\widehat{\Delta R}$ vs. CE (calibration-matched)		
Ratio	Seed	CE	CB	DRW	LA	CB	DRW	LA
10	0	0.5388	0.5154	0.5447	0.5497	−0.06903	+0.00617	+0.02222
50	0	0.4256	0.3242	0.4399	0.4550	−0.24409	+0.01903	+0.03768
100	0	0.3662	0.2627	0.3874	0.3959	−0.23936	+0.01987	+0.03449
100	1	0.3614	0.2287	0.3762	0.3877	−0.26176	+0.00748	+0.02672
100	2	0.3687	0.2481	0.3968	0.4024	−0.25213	+0.03124	+0.03601
100, mean (sd)						−0.25108 (0.01124)	+0.01953 (0.01188)	+0.03241 (0.00498)

roughly fourfold across seeds at ratio 100 (+0.007 to +0.031). We therefore report the spread rather than a seed-level interval: the direction replicates, the magnitude is seed-sensitive, and the gap between the tight within-run intervals of Table 9 and this across-seed spread is exactly why a method-level claim needs replication rather than a single audit.

Third, LA is an instructive falsification of a natural expectation. A purely post-hoc prior correction might be expected to register as almost pure prior channel, as the within-cell prior matching of Section 4.6 does. It does not: calibration-matched, LA reads as almost entirely alignment (+0.022 to +0.038) with a prior channel near zero, while achieving the best balanced accuracy of any arm at every ratio. The reason is visible in the construction. Dividing by the *global* class prior is not a within-superclass histogram move, because fine classes inside a superclass differ in training frequency, and the correction it applies is multiplicative in each example's own predicted probabilities — it changes rankings where the model already carried latent evidence. The two post-hoc corrections in this paper therefore land in opposite channels, and correctly so: transport separates adjustments that act on the cell's label histogram from adjustments that act on individual predictions.

Where the alignment gain sits is visible in Table 10 and Figure 7 (raw regime, primary seed). For DRW it falls where the long-tail literature intends it to, reaching +0.09009 on few-shot classes and +0.06760 on medium-shot ones, while many-shot classes lose −0.07624 as the schedule shifts capacity away from the head. The class-balanced run reproduces that same ordering with every level displaced downward, its many-shot alignment collapsing to −0.41977. What the decomposition supplies, then, is not only how much of a long-tail gain is real but where in the frequency spectrum it was earned.

These numbers also provide an implementation check on real inputs. Computing the alignment term directly from the covariance identity of Eq. (7), through a code path that shares nothing with the transport estimator, gives −0.19530; multiplying by the factor $n_c/(n_c - 1)$ that Proposition 3 predicts, at the $n_c = 500$ of a balanced superclass cell, returns −0.195693 against the estimator's −0.19569. The identities of Theorem 1 and Proposition 3 are algebraic, so this verifies the implementation rather than the theory; the point of the check is that two independent code paths agree to the fifth decimal on real trained models, not only on the synthetic populations used in the unit

Table 9: WAGER decomposition on the 10,000-image balanced CIFAR-100-LT test split. Gains use the quadratic score; CI is the Hájek interval on $\widehat{\Delta R}$; p is the 499-shift randomization diagnostic, one-sided for positive alignment, so the value of 1.000 on the CB rows records an alignment gain at the extreme negative end of its randomization distribution.

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (20)	0.00143	0.19713	−0.19569 [−0.20728, −0.18411]	1.000
CB vs CE	global (1)	0.00143	0.25573	−0.25430 [−0.26801, −0.24058]	1.000
CB vs CE	tier (3)	0.00143	0.25247	−0.25103 [−0.26453, −0.23754]	1.000
DRW vs CE	superclass (20)	0.06606	0.04206	0.02400 [0.01340, 0.03460]	.002
DRW vs CE	global (1)	0.06606	0.03522	0.03085 [0.01860, 0.04309]	.002
DRW vs CE	tier (3)	0.06606	0.03702	0.02904 [0.01702, 0.04107]	.002

Table 10: Alignment gain by training-frequency tier at the primary superclass audit. The deferred schedule concentrates its genuine per-instance improvement on rare classes; both methods lose alignment on the head.

		CB vs CE	DRW vs CE	
Tier	n	$\widehat{\Delta R}$	$\widehat{\Delta T}$	$\widehat{\Delta R}$
Few-shot (< 20)	3000	−0.03337	0.13785	0.09009
Medium-shot (20–100)	3500	−0.11075	0.10799	0.06760
Many-shot (> 100)	3500	−0.41977	−0.03740	−0.07624

tests.

Coarsening behaves as Proposition 1 predicts. Moving from the 20 superclass cells to a single global cell changes the credited alignment from −0.19569 to −0.25430; the difference is the between-cell covariance term, which is negative here, so the coarse audit overstates the alignment loss by attributing to it structure that the superclass partition correctly assigns to ΔP. The finer partition is again the conservative reading, for the same reason as in Section 4.4 and now in a second domain.

4.8. Cross-domain validation: long-tailed text classification

Sections 4.3–4.7 test two vision tasks that share cached probability vectors and audit features grounded in class frequency or object geometry. To test the breadth Section 5.2 claims – any benchmark supplying two models' full probability vectors, labels, and a declared discrete prior feature – we apply the identical estimator, unmodified, to document category prediction on 20 Newsgroups [32]. Neither the predictors nor the audit feature are visual here: documents replace images, and the declared ϕ is a topical grouping rather than a spatial or class-frequency one, so a favorable result cannot be attributed to anything specific to vision data.

We build a long-tailed training subset with the same exponential profile as CIFAR-100-LT (imbalance ratio 100, Section 4.7), applied to 20 Newsgroups' 20 fine categories, keeping the standard balanced test split (7,532 documents) untouched (Appendix Appendix D). The 20 categories fall into six conventional coarse topics – computing, recreation, science, politics, religion, and classifieds – which we use as the

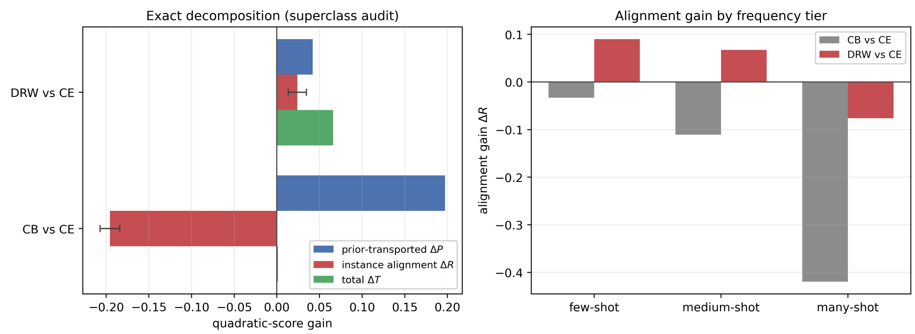

Figure 7: **The same attribution transfers to a second task.** Left: exact decomposition at the superclass audit. The CB total gain (green) is visually almost absent at +0.00143 while its two channels reach ±0.2 – a near-zero aggregate score concealing two large opposing effects. DRW's smaller total is genuinely split between both channels. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: the alignment channel by training-frequency tier, showing that DRW's genuine per-instance gain is concentrated on rare classes and negative on the head.

primary declared ϕ, the direct analogue of CIFAR's coarse superclasses; a global (trivial) and a training-frequency-tier partition are reported alongside, as in Section 4.7. Documents are represented by a TF-IDF vector reduced to 300 dimensions by truncated SVD, and the three compared classifiers – CE, CB, and DRW – follow the same three-arm protocol as Section 4.7: an identical small MLP architecture and optimizer across runs, differing only in per-example loss weight (Appendix Appendix D). Test documents are independent units, so the plain Hájek interval of Section 3.9 applies directly, as for CIFAR.

Result.. Table 11 decomposes both re-weighting schedules' gain over the cross-entropy baseline (top-1 accuracies 0.3532 CE, 0.3960 CB, 0.3776 DRW). The two schedules land in opposite channels here, which is itself the finding worth reporting: on this task it is class-balanced re-weighting from initialization, not the deferred schedule, that buys genuine within-topic instance alignment (+0.04317, CI $[0.03664, 0.04970]$, 37.9% of its total gain), while DRW's comparably sized total gain (0.11490 vs. CB's 0.11388) is 98.6% prior-transported, its residual alignment statistically indistinguishable from zero at the superclass audit (+0.00155, CI $[-0.00296, 0.00605]$, $p = .236$) and significantly negative once the coarser tier partition is used instead (−0.00901, CI $[-0.01428, -0.00374]$). This is the reverse of Section 4.7's image-classification finding, where CB's near-zero total masked a large alignment *loss* and DRW's gain was genuinely alignment-driven. Nothing about the estimator or the construction of ϕ changed between the two applications; what changed is which training recipe, on which data, buys real per-instance discrimination – exactly the composition question an aggregate score or an accuracy difference alone cannot answer, now demonstrated on a domain with neither a class-frequency table nor a spatial prior.

Coarsening moves both comparisons in the direction Proposition 1 allows without requiring: CB's credited alignment falls from 0.04317 at the superclass partition to 0.04185 at the global cell and 0.03578 at the tier partition, while DRW's changes sign

Table 11: WAGER decomposition on the 7,532-document balanced 20-Newsgroups-LT test split. Gains use the quadratic score; CI is the (unclustered) Hájek interval on $\widehat{\Delta R}$, documents being independent units; p is the 499-shift randomization diagnostic, one-sided for positive alignment.

Comparison	Audit cell ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ (95% CI)	p
CB vs CE	superclass (6)	0.11388	0.07071	0.04317 [0.03664, 0.04970]	.002
CB vs CE	global (1)	0.11388	0.07203	0.04185 [0.03381, 0.04990]	.002
CB vs CE	tier (3)	0.11388	0.07810	0.03578 [0.02825, 0.04331]	.002
DRW vs CE	superclass (6)	0.11490	0.11335	0.00155 [−0.00296, 0.00605]	.236
DRW vs CE	global (1)	0.11490	0.11809	−0.00319 [−0.00884, 0.00245]	.980
DRW vs CE	tier (3)	0.11490	0.12391	−0.00901 [−0.01428, −0.00374]	1.000

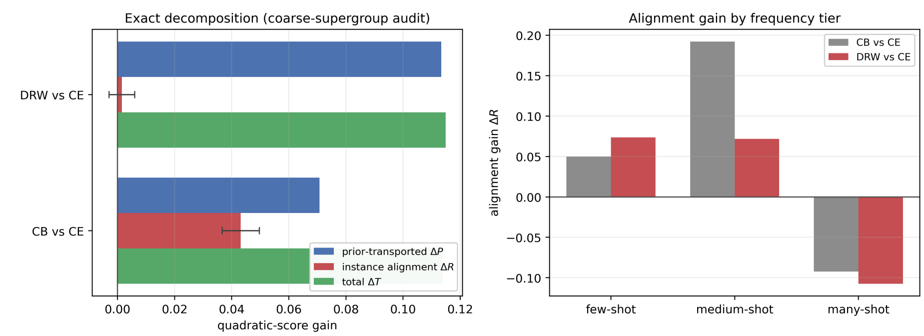

Figure 8: **A third domain, with neither a spatial nor a class-frequency prior.** Left: exact decomposition at the coarse-topic audit; CB's total gain is genuinely split between both channels while DRW's is almost entirely prior-transported. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: alignment gain by training-frequency tier for both schedules, showing the same head-loses/tail-gains signature as Figure 7 despite the change of domain and audit feature.

twice, from a small positive residual (superclass) to a small negative one (global) to a significant negative one (tier). None of these differences overturns CB's conclusion, but DRW's illustrates exactly why the finest defensible partition is the conservative default (Section 3.4): a reader stopping at the coarser tier partition would report a significant alignment *loss* for a method whose finer-grained residual is not distinguishable from zero.

Table 12 breaks the superclass-audit alignment down by training-frequency tier for both schedules; the two methods agree in shape even though their superclass totals diverge sharply in composition, each gaining alignment on rare and medium-frequency categories and losing it on frequent ones, the same qualitative signature Section 4.7 reports for image classification.

This third domain uses a single trained pair per schedule rather than the multi-seed, calibration-matched protocol Section 4.7 applies to CIFAR-100-LT; extending that fuller protocol to text is the natural next replication, not a step this section claims to have taken.

Table 12: Alignment gain by training-frequency tier at the superclass audit, 20-Newsgroups-LT. Both schedules gain instance alignment on rare and medium categories and lose it on frequent ones.

Tier	n	$\widehat{\Delta R}$, CB vs CE	$\widehat{\Delta R}$, DRW vs CE
Few-shot (< 20)	1954	0.04991	0.07352
Medium-shot (20–100)	2610	0.19213	0.07171
Many-shot (> 100)	2968	−0.09227	−0.10753

5. Discussion and Limitations

5.1. What the decomposition does and does not establish

WAGER changes the unit of analysis from a model to a model-pair gain. This matters scientifically. A class-only network may be a better prior estimator than FREQ, but that fact does not become evidence of instance-specific visual use. Conversely, a model can improve visual alignment while losing enough prior score to obscure the improvement in its aggregate metric. The signed identity $\Delta T = \Delta P + \Delta R$ exposes both cases without a separately fitted reference forecaster.

The interpretation remains conditional on the declared audit feature ϕ. A positive ΔR means the new model aligns its probability changes with the correct example better than the old model among examples sharing ϕ. It does not prove causal inference, compositional understanding, or robustness. If an unrecorded shortcut varies inside a cell, WAGER will count its predictive use in ΔR; Proposition 2 turns this qualitative caveat into a numeric one, bounding how far the reported alignment gain could be from what a fully adjusted audit would find as a function of a single, elicitable correlation parameter $\bar{\rho}$, though Table 1 shows that bound is conservative at VG150's label cardinality. Applications should therefore choose ϕ before looking at results, justify it from the benchmark's data generation process, and report finer and coarser alternatives. The observed increase under subject-only cells illustrates Proposition 1: coarsening adds a between-cell covariance term of either sign, so it cannot be assumed to leave the estimate unchanged, and only the finest identified partition isolates alignment gain net of every prior regularity expressible in ϕ. This is why the finest defensible prior feature is the conservative primary choice, not merely an empirical convention.

Two further interpretation limits deserve explicit statement. First, ΔR is not invariant to monotone recalibration: because the alignment channel is a covariance between probability movements and labels, sharpening or softening a fixed model shifts score mass between the two channels while adding no information about any individual example. Section 4.7 quantifies the effect on real models and adopts the corresponding control – decompose calibration-matched model pairs, with temperatures chosen on data disjoint from the audit – which we recommend whenever the compared systems differ visibly in confidence. Second, a prior-transported gain is not evidence of anything illegitimate. Fitting the deployment label distribution better is genuine, decision-relevant skill whenever that distribution matches the benchmark's; it is fragile precisely under label shift, where prior-transported gain can be added or removed post hoc by re-weighting [45, 36], as the prior-matching experiment of Section 4.6 demonstrates directly. The decomposition reports where a gain lives; whether the prior channel is worth paying for is a property of the deployment, not of the model pair.

"Choose ϕ before looking" is stated advice, not an enforced mechanism, and the analyst-chosen-subgroup problem has structural answers that WAGER should inherit. Our recommendation is that the *benchmark*, not the submitting author, declare the audit feature – the annotation prior is a property of the dataset, and maintainer-declared ϕ removes the incentive to report only the most favorable of several candidates. Where several defensible priors exist, all pre-declared candidates should be reported, as our sensitivity tables do; guarantees that hold uniformly over a class of groupings are the subject of multicalibration [23], and porting that style of guarantee to pairwise gain attribution is a natural extension. To make reports comparable across papers, we suggest a fixed reporting block per audited pair: the declared ϕ and its rationale, the identified coverage, both channels signed with the alignment interval, the calibration-matched row when the models differ visibly in confidence, and the granularity sensitivity rows.

Exact transport also has a positivity requirement: a cell needs at least two examples. WAGER reports the identified fraction and makes no claim for singletons. Very fine ϕ can therefore improve confound control while reducing coverage. One further subtlety: cells identified at a fine granularity need not coincide with those at a coarse one, so cross-granularity comparisons mix the between-cell covariance term of Proposition 1 with a change of identified subpopulation – immaterial at VG150's 99.0% coverage, but worth restricting to a common identified subsample when coverage is low. Approximate matching could extend the method to continuous priors, but would replace the exact identity's clear intervention with modeling assumptions; we leave that extension open.

The image-clustered confidence interval is asymptotic, and Theorem 4 is what justifies that: under a bounded score and a mild condition ruling out one cluster from dominating the sample, the dataset-level estimator is asymptotically normal around the identified-subpopulation estimand, and the same sandwich variance already reported is consistent for its limiting variance. The cyclic randomization test is finite-group exact only under its sharper within-cell exchangeability null and treats relations as the randomized units, so it is secondary in the SGG analysis. A future image-level randomization design that preserves several relation-cell margins simultaneously would strengthen finite-sample inference for structured benchmarks.

The audit of Section 4.5 indicates what adoption actually requires. Two released checkpoints, evaluated with their authors' own code, were enough: the method consumed the resulting probability arrays and returned a decomposition with intervals in seconds. Nothing about that workflow is specific to scene graph generation, and the practical obstacle is neither compute nor complexity but disclosure — benchmarks overwhelmingly collect ranked predictions rather than predictive distributions. A benchmark that archived per-example probabilities alongside its leaderboard would make this kind of audit routine for every submitted pair. Where several audit features are defensible, we would further recommend that the benchmark, rather than the submitting author, declare ϕ, and that all pre-declared candidates be reported.

Finally, WAGER evaluates probability vectors, not only top-1 decisions. This is a feature—the method can detect useful redistribution below the argmax—but it requires released logits or probabilities. The quadratic score is bounded and strictly proper;

other proper scores answer scale-dependent variants of the same question and should be declared in advance.

5.2. Broader applicability

Four aligned arrays are all the algorithm requires – two models' probability vectors, the labels, and the prior-cell identifiers – and nothing in Sections 3.1–3.9 is specific to relation prediction. Section 4.7 puts that generality to the test on long-tailed image classification, where taking the benchmark's coarse superclasses as ϕ lets the decomposition distinguish a re-weighting schedule whose near-zero aggregate score hides two large cancelling channels from one whose genuine improvement is part prior-refitting and part rare-class alignment. The claim of breadth thereby acquires empirical content instead of resting on analogy. That second domain is also where the constraint on ϕ shows its teeth, since the fine class label cannot serve as an audit cell: it would leave every cell label-homogeneous and drive ΔR to zero by construction, producing an artifact that reads exactly like a null result. Elsewhere the natural choices are easy enough to name – a predeclared question template for VQA, an answer format or prompt family for other multimodal benchmarks – though we have not evaluated them, and we flag them as the obvious next domains rather than claim a coverage we have not earned. Whatever the setting, the scientific claim has to stay precise, because what WAGER measures is new prediction–label alignment within the declared cells and nothing more. Its value lies not in any single ϕ settling what counts as understanding, but in making every attribution an explicit and reproducible counterfactual tied to a declared nuisance structure, one that the same unmodified construction has now answered sensibly on two structurally different tasks.

6. Conclusion

We introduced WAGER, Within-cell Antisymmetric Gain Evaluation of Resolution, a direct algorithm for attributing the score improvement between two frozen probabilistic classifiers. By crossing labels among examples with the same prior feature, WAGER constructs the prior-only counterfactual from the evaluated data themselves and obtains an exact identity between total, prior-transported, and instance-alignment gains. The residual is an interpretable U-statistic that, under the quadratic score, equals a conditional covariance gain, and the same covariance identity holds for the entire Bregman-score family (Theorem 2) rather than the quadratic score alone. The estimator is computable in $O(NK)$ time, its dataset-level statistic is asymptotically normal around the identified-subpopulation estimand under conditions that hold whenever the score is bounded and no single cluster dominates the sample (Theorem 4), and it reduces, at a trivial audit feature, to a clustered Diebold–Mariano/Giacomini–White test (Corollary 3) – so every real-data comparison in this paper already reports what that classical test would, plus the split it does not provide. A Cauchy–Schwarz sensitivity bound (Proposition 2) turns the paper's central qualitative caveat – that an unrecorded confounder can masquerade as alignment – into a numeric limit on how far the reported residual could be from a fully adjusted audit, elicited from a single correlation parameter rather than a model of the confounder itself. Controlled experiments support null calibration, signal recovery, and the coverage of the reported intervals, and establish what the two channels do and do not separate: prior-only improvements stay out of the alignment channel, unrecorded within-cell shortcuts enter it, and a monotone recalibration moves score between the channels, which is why calibration-matched auditing is part of the protocol rather than an afterthought.

Pointed at two released scene-graph checkpoints rather than at predictors of our own, the method returns a precise account of a widely used causal debiasing technique: its ten-point mean-recall improvement is accompanied, once calibration is matched, by an instance-alignment change indistinguishable from zero, with essentially the entire proper-score difference lying in the prior channel. That is what its construction subtracts, so the finding characterizes the method rather than indicting it – and it shows that a metric the field reads as evidence about rare predicates can move ten points without any measurable change in instance-level evidence.

On our own Visual Genome predictors the method assigns a class-only improvement entirely to the prior channel while identifying the additional alignment that spatial geometry supplies. Substituting a frozen CLIP encoder for that geometry produces the sharpest illustration of why the two channels are worth separating: the resulting model scores and ranks worse by every aggregate measure, yet is shown to align with individual instances significantly better than the model it appears to lose to, its whole deficit lying in the annotation prior and its advantage surviving both post-hoc prior correction and calibration matching. Carried over without modification to long-tailed classification, the estimator exposes the composition of two re-weighting schedules' gains, and the calibration-matched reading overturns what the raw scores suggest about each: the schedule whose aggregate barely moves has genuinely lost within-class discrimination, while the one that improves has bought almost purely instance alignment rather than the prior refitting its raw decomposition implies. Neither an aggregate score nor an accuracy difference can make that separation.

A third domain, long-tailed text classification on 20 Newsgroups (Section 4.8), carries no spatial or class-frequency prior at all, and supplies the estimator's sharpest demonstration that channel composition is a property of a training recipe applied to particular data, not of the recipe's name: on this task it is the class-balanced schedule that buys genuine instance alignment, and the deferred schedule – the one that carried the real alignment gain on CIFAR-100-LT – whose comparably large total gain is almost entirely prior-transported. The two vision domains and the one text domain are audited by the identical unmodified construction, differing only in what ϕ is declared to be. It is this gain-specific mechanism, together with its extension to a general score family, its large-sample justification, its explicit sensitivity to what it cannot see, and its exact relation to classical forecast-comparison tests, rather than the familiar combination of a frequency baseline with a generic significance test, that constitutes the paper's technical contribution.

Data and code availability

All code, cached model outputs and scripts needed to reproduce every number, table and figure in this paper are publicly available at `https://github.com/vinhqdang/WAGER`. The repository contains the estimator, its unit tests, every experiment driver, and the cached results files that each reported value is checked against by an included verification script. The two audited scene-graph checkpoints are the public release of Tang et al. [49]; Visual Genome [30] and CIFAR-100 are public benchmarks.

Declaration of competing interests

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used a generative AI assistant in order to polish the text and correct grammar. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

[1] Agrawal, A., Batra, D., Parikh, D., Kembhavi, A., 2018. Don't just assume; look and answer: Overcoming priors for visual question answering, in: CVPR, pp. 4971–4980. doi:10.1109/CVPR.2018.00522.

[2] Banerjee, B., Bhattacharya, B.B., Ghosh, A.K., 2026. A ball divergence-based measure for conditional independence testing with a local wild bootstrap. Biometrika 113. doi:10.1093/biomet/asag027.

[3] Berrett, T.B., Wang, Y., Barber, R.F., Samworth, R.J., 2020. The conditional permutation test for independence while controlling for confounders. Journal of the Royal Statistical Society: Series B 82, 175–197. doi:10.1111/rssb.12340.

[4] Bröcker, J., 2009. Reliability, sufficiency, and the decomposition of proper scores. Quarterly Journal of the Royal Meteorological Society 135, 1512–1519. doi:10.1002/qj.456.

[5] Brown, E., Yang, J., Yang, S., Fergus, R., Xie, S., 2025. Benchmark designers should "train on the test set" to expose exploitable non-visual shortcuts. arXiv preprint arXiv:2511.04655 doi:10.48550/arXiv.2511.04655.

[6] Cai, L., Hu, Q., Li, W., 2026. Asymptotic anytime-valid inference for u-statistics. arXiv preprint arXiv:2605.14692 doi:10.48550/arXiv.2605.14692.

[7] Cao, K., Wei, C., Gaidon, A., Arechiga, N., Ma, T., 2019. Learning imbalanced datasets with label-distribution-aware margin loss, in: NeurIPS. doi:10.48550/arXiv.1906.07413.

[8] Charpentier, A., Fernandes Machado, A., 2026. Decomposing probabilistic scores: Reliability, information loss and uncertainty. arXiv preprint arXiv:2603.15232 doi:10.48550/arXiv.2603.15232.

[9] Chen, T., Yu, W., Chen, R., Lin, L., 2019. Knowledge-embedded routing network for scene graph generation, in: CVPR, pp. 6163–6171. doi:10.1109/CVPR.2019.00632.

[10] Cinelli, C., Hazlett, C., 2020. Making sense of sensitivity: Extending omitted variable bias. Journal of the Royal Statistical Society: Series B 82, 39–67. doi:10.1111/rssb.12348.

[11] Cui, Y., Jia, M., Lin, T.Y., Song, Y., Belongie, S., 2019. Class-balanced loss based on effective number of samples, in: CVPR, pp. 9268–9277. doi:10.1109/CVPR.2019.00949.

[12] DeGroot, M.H., Fienberg, S.E., 1983. The comparison and evaluation of forecasters. Journal of the Royal Statistical Society: Series D (The Statistician) 32, 12–22. doi:10.2307/2987588.

[13] Diebold, F.X., Mariano, R.S., 1995. Comparing predictive accuracy. Journal of Business & Economic Statistics 13, 253–263. doi:10.1080/07350015.1995.10524599.

[14] Duan, C.L., Li, Y., Wei, X.S., Zhao, L., 2024. Long-tailed object detection pretraining: Dynamic rebalancing contrastive learning with dual reconstruction, in: NeurIPS, pp. 40230–40253. doi:10.52202/079017-1272.

[15] Ethayarajh, K., Choi, Y., Swayamdipta, S., 2022. Understanding dataset difficulty with v-usable information, in: ICML, PMLR. pp. 5988–6008. doi:10.48550/arXiv.2110.08420.

[16] Freidling, T., Zhao, Q., 2025. Optimization-based sensitivity analysis for unmeasured confounding using partial correlations. Journal of Computational and Graphical Statistics doi:10.1080/10618600.2025.2573156.

[17] Geirhos, R., Jacobsen, J.H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., Wichmann, F.A., 2020. Shortcut learning in deep neural networks. Nature Machine Intelligence 2, 665–673. doi:10.1038/s42256-020-00257-z.

[18] Giacomini, R., White, H., 2006. Tests of conditional predictive ability. Econometrica 74, 1545–1578. doi:10.1111/j.1468-0262.2006.00718.x.

[19] Gneiting, T., Raftery, A.E., 2007. Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association 102, 359–378. doi:10.1198/016214506000001437.

[20] Goyal, Y., Khot, T., Summers-Stay, D., Batra, D., Parikh, D., 2017. Making the v in vqa matter: Elevating the role of image understanding in visual question answering, in: CVPR, pp. 6904–6913. doi:10.1109/CVPR.2017.670.

[21] He, C., Gao, Y., Zhu, L., Huang, J., 2025. From conditional to unconditional independence: Testing conditional independence via transport maps. arXiv preprint arXiv:2504.09567 doi:10.48550/arXiv.2504.09567.

[22] He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition, in: CVPR, pp. 770–778. doi:10.1109/CVPR.2016.90.

[23] Hébert-Johnson, Ú., Kim, M.P., Reingold, O., Rothblum, G.N., 2018. Multicalibration: Calibration for the (computationally-identifiable) masses, in: ICML, pp. 1939–1948. doi:10.48550/arXiv.1711.08513.

[24] Hewitt, J., Ethayarajh, K., Liang, P., Manning, C.D., 2021. Conditional probing: measuring usable information beyond a baseline, in: EMNLP, pp. 1626–1639. doi:10.18653/v1/2021.emnlp-main.122.

[25] Hoeffding, W., 1948. A class of statistics with asymptotically normal distribution. The Annals of Mathematical Statistics 19, 293–325. doi:10.1214/aoms/1177730196.

[26] Hofman, P., Sale, Y., Hüllermeier, E., 2025. Uncertainty quantification with proper scoring rules: Adjusting measures to prediction tasks. arXiv preprint arXiv:2505.22538 doi:10.48550/arXiv.2505.22538.

[27] Kang, B., Xie, S., Rohrbach, M., Yan, Z., Gordo, A., Feng, J., Kalantidis, Y., 2020. Decoupling representation and classifier for long-tailed recognition, in: ICLR. doi:10.48550/arXiv.1910.09217.

[28] Kervadec, C., Antipov, G., Baccouche, M., Wolf, C., 2021. Roses are red, violets are blue... but should vqa expect them to?, in: CVPR, pp. 2775–2784. doi:10.1109/CVPR46437.2021.00280.

[29] Kim, Y., Kim, I., 2026. When do generalized permutation tests achieve optimal power? a dispersion characterization. arXiv preprint arXiv:2606.00578 doi:10.48550/arXiv.2606.00578.

[30] Krishna, R., Zhu, Y., Groth, O., Johnson, J., Hata, K., Kravitz, J., Chen, S., Kalantidis, Y., Li, L.J., Shamma, D.A., et al., 2017. Visual genome: Connecting language and vision using crowdsourced dense image annotations. International Journal of Computer Vision 123, 32–73. doi:10.1007/s11263-016-0981-7.

[31] Kuang, P., Wang, Z., Chu, Z., Wang, J., Ren, K., 2024. Towards real-world debiasing: Rethinking evaluation, challenge, and solution. arXiv preprint arXiv:2405.15240 doi:10.48550/arXiv.2405.15240.

[32] Lang, K., 1995. Newsweeder: Learning to filter netnews, in: ICML, pp. 331–339. doi:10.1016/B978-1-55860-377-6.50048-7.

[33] Li, M., Hu, Z., Lu, Y., Lan, W., Cheung, Y.m., Huang, H., 2026. Pi-h2t: Enhancing long-tailed visual recognition with permutation-invariant and head-to-tail feature fusion. IEEE Transactions on Pattern Analysis and Machine Intelligence doi:10.1109/TPAMI.2026.3711911.

[34] Li, R., Zhang, S., Lin, D., Chen, K., He, X., 2024. From pixels to graphs: Open-vocabulary scene graph generation with vision-language models, in: CVPR. doi:10.1109/CVPR52733.2024.02652.

[35] Li, W., Zhang, H., Bai, Q., Zhao, G., Jiang, N., Yuan, X., 2022. Rethinking the evaluation of unbiased scene graph generation, in: BMVC. doi:10.48550/arXiv.2208.01909.

[36] Lipton, Z.C., Wang, Y.X., Smola, A., 2018. Detecting and correcting for label shift with black box predictors, in: ICML, pp. 3122–3130. doi:10.48550/arXiv.1802.03916.

[37] Lu, C., Krishna, R., Bernstein, M., Fei-Fei, L., 2016. Visual relationship detection with language priors, in: ECCV, pp. 852–869. doi:10.1007/978-3-319-46448-0_51.

[38] Luo, J., Hong, F., Hu, Q., Cao, X., Liu, F., Yao, J., 2025. Long-tailed recognition with model rebalancing, in: NeurIPS. doi:10.48550/arXiv.2510.08177.

[39] Ma, Y., Jiao, L., Liu, F., Yang, S., Liu, X., Chen, P., 2024. Geometric prior guided feature representation learning for long-tailed classification. International Journal of Computer Vision doi:10.1007/s11263-024-01983-2.

[40] Menon, A.K., Jayasumana, S., Rawat, A.S., Jain, H., Veit, A., Kumar, S., 2021. Long-tail learning via logit adjustment, in: ICLR. doi:10.48550/arXiv.2007.07314.

[41] Murphy, A.H., 1973. A new vector partition of the probability score. Journal of Applied Meteorology 12, 595–600. doi:10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2.

[42] Ojala, M., Garriga, G.C., 2010. Permutation tests for studying classifier performance. Journal of Machine Learning Research 11, 1833–1863.

[43] Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I., 2021. Learning transferable visual models from natural language supervision, in: ICML, pp. 8748–8763. doi:10.48550/arXiv.2103.00020.

[44] Rosenbaum, P.R., 2002. Observational Studies. Springer Series in Statistics. 2 ed., Springer, New York. doi:10.1007/978-1-4757-3692-2.

[45] Saerens, M., Latinne, P., Decaestecker, C., 2002. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. Neural Computation 14, 21–41. doi:10.1162/089976602753284446.

[46] Saha, A., Ramdas, A., 2024. Testing exchangeability by pairwise betting, in: Proceedings of the 27th International Conference on Artificial Intelligence and Statistics, PMLR, Valencia, Spain. pp. 4915–4923. doi:10.48550/arXiv.2310.14293.

[47] Serfling, R.J., 1980. Approximation Theorems of Mathematical Statistics. Wiley Series in Probability and Statistics, John Wiley & Sons, New York.

[48] Sun, G., Qiu, J., Liang, L., 2024. Unbiased scene graph generation by type-aware message passing on heterogeneous and dual graphs. arXiv preprint arXiv:2411.13287 doi:10.48550/arXiv.2411.13287.

[49] Tang, K., Niu, Y., Huang, J., Shi, J., Zhang, H., 2020. Unbiased scene graph generation from biased training, in: CVPR, pp. 3716–3725. doi:10.1109/CVPR42600.2020.00377.

[50] Tang, K., Zhang, H., Wu, B., Luo, W., Liu, W., 2019. Learning to compose dynamic tree structures for visual contexts, in: CVPR, pp. 6619–6628. doi:10.1109/CVPR.2019.00678.

[51] Teney, D., Kafle, K., Shrestha, R., Abbasnejad, E., Kanan, C., van den Hengel, A., 2020. On the value of out-of-distribution testing: An example of goodhart's law, in: NeurIPS. doi:10.48550/arXiv.2005.09241.

[52] Torralba, A., Efros, A.A., 2011. Unbiased look at dataset bias, in: CVPR, pp. 1521–1528. doi:10.1109/CVPR.2011.5995347.

[53] van der Vaart, A.W., 1998. Asymptotic Statistics. volume 3 of *Cambridge Series in Statistical and Probabilistic Mathematics*. Cambridge University Press, Cambridge, UK.

[54] Vieira, B.H., 2026. An intuitive rearranging of the yates covariance decomposition for probabilistic verification of forecasts with the brier score. arXiv preprint arXiv:2603.05544 doi:10.48550/arXiv.2603.05544.

[55] Waghmare, K., Ziegel, J., 2026. Proper scoring rules for estimation and forecast evaluation. Annual Review of Statistics and Its Application 13. doi:10.1146/annurev-statistics-042424-050626.

[56] Xu, D., Zhu, Y., Choy, C.B., Fei-Fei, L., 2017. Scene graph generation by iterative message passing, in: CVPR, pp. 3097–3106. doi:10.1109/CVPR.2017.330.

[57] Xu, Q., Zhou, L., Zhong, X., Zhang, F., Huang, R., Lin, C.W., 2025. Qirl: Boosting visual question answering via optimized question-image relation learning. arXiv preprint arXiv:2504.03337 doi:10.48550/arXiv.2504.03337.

[58] Xu, Y., Zhao, S., Song, J., Stewart, R., Ermon, S., 2020. A theory of usable information under computational constraints, in: ICLR. doi:10.48550/arXiv.2002.10689.

[59] Zellers, R., Yatskar, M., Thomson, S., Choi, Y., 2018. Neural motifs: Scene graph parsing with global context, in: CVPR, pp. 5831–5840. doi:10.1109/CVPR.2018.00611.

[60] Zhang, C., Stepputtis, S., Campbell, J., Sycara, K., Xie, Y., 2024. Hiker-sgg: Hierarchical knowledge enhanced robust scene graph generation, in: CVPR. doi:10.1109/CVPR52733.2024.02667.

[61] Zhang, Y., Zhao, Q., 2026. An average-case sensitivity analysis for unmeasured confounding. Biometrika 113. doi:10.48550/arXiv.2211.04697.

[62] Zhou, W., Liu, F., Zheng, H., Zhao, R., 2025. Mitigating data bias and ensuring reliable evaluation of ai models with shortcut hull learning. Nature Communications 16. doi:10.1038/s41467-025-60801-6.

Appendix A. Proofs

Appendix A.1. Proof of Theorem 1

Condition throughout on $C = c$. Expanding over labels gives

$$\Delta T_c = \sum_y \mathbb{E}[H_X(y)\mathbb{1}\{Y = y\}], \qquad \Delta P_c = \sum_y \mathbb{E}[H_X(y)]\Pr(Y = y).$$

Subtracting yields the covariance sum in Eq. (6); the first identity is the definition of ΔR_c. Under the quadratic score,

$$H_X(y) = 2\Delta q_y(X) - \{\|q_1(\cdot \mid X)\|_2^2 - \|q_0(\cdot \mid X)\|_2^2\}.$$

The bracketed term is independent of the label index. Its contribution summed over y is $\mathrm{Cov}(B(X), \sum_y \mathbb{1}\{Y = y\}) = \mathrm{Cov}(B(X), 1) = 0$. The remaining terms give Eq. (7). If $H_X(y) = H_c(y)$ almost surely, every covariance is zero. □

Appendix A.2. Proof of Theorem 2

Condition on $C = c$. The proof of Theorem 1 above establishes $\Delta R_c = \sum_y \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c)$ for *any* score contrast H, not only the quadratic one; only the identification of $H_x(y)$ with a specific functional form is score-dependent. For the Bregman score of Eq. (8),

$$H_x(y) = \underbrace{[\psi(q_1(x)) - \psi(q_0(x))] - [\langle\nabla\psi(q_1(x)), q_1(x)\rangle - \langle\nabla\psi(q_0(x)), q_0(x)\rangle]}_{=:B(x)} + \Delta g_y(x),$$

writing $q_i(x)$ for $q_i(\cdot \mid x)$, where $B(x)$ collects every term not indexed by y and $\Delta g_y(x) = \partial_y\psi(q_1(x)) - \partial_y\psi(q_0(x))$ is the y-indexed remainder, exactly as in the proof of Theorem 1 for the quadratic case. Substituting,

$$\sum_y \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid c) = \mathrm{Cov}\Big(B(X), \sum_y \mathbb{1}\{Y = y\} \,\Big|\, c\Big) + \sum_y \mathrm{Cov}(\Delta g_y(X), \mathbb{1}\{Y = y\} \mid c).$$

The first term is $\mathrm{Cov}(B(X), 1 \mid c) = 0$ since $\sum_y \mathbb{1}\{Y = y\} = 1$ almost surely. The second term is Eq. (10). □

Appendix A.3. Proof of Theorem 3

Summing the kernel in Eq. (20) over unordered pairs, every observed term $H_i(y_i)$ occurs in $n_c - 1$ pairs with coefficient $1/2$, while the two crossed terms together enumerate every ordered pair $H_i(y_j)$, $i \neq j$, with coefficient $1/2$. Since $\binom{n_c}{2} = n_c(n_c - 1)/2$,

$$\widehat{\Delta R}_c = \frac{1}{n_c} \sum_i H_i(y_i) - \frac{1}{n_c(n_c - 1)} \sum_{i \neq j} H_i(y_j) = \widehat{\Delta T}_c - \widehat{\Delta P}_c.$$

This is deterministic. When the cell's examples are i.i.d. draws from the cell population, the average of the symmetric order-two kernel is the standard unbiased U-statistic for its population expectation [25]; independence is used, not merely exchangeability, because ΔP_c is defined through an independent label coupling, so $\mathbb{E}[H_1(Y_2)] = \mathbb{E}[H_X(Y')]$ requires the two draws to be independent. □

Appendix A.4. Proof of Proposition 1

Fix y and condition throughout on $\bar{\phi} = \bar{c}$. The law of total covariance applied to the pair $(H_X(y), \mathbb{1}\{Y = y\})$ with the finer conditioning variable C nested inside $\bar{c}$ gives

$$\begin{aligned}\operatorname{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid \bar{\phi} = \bar{c}) = \mathbb{E}\big[&\operatorname{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C) \,\big|\, \bar{\phi} = \bar{c}\big] \\ &+ \operatorname{Cov}\big(\mathbb{E}[H_X(y) \mid C], \Pr(Y = y \mid C) \,\big|\, \bar{\phi} = \bar{c}\big),\end{aligned}$$

using $\mathbb{E}[\mathbb{1}\{Y = y\} \mid C] = \Pr(Y = y \mid C)$. Summing over $y = 1, \dots, K$ and applying the population identity of Theorem 1 to the left side (at $\bar{\phi} = \bar{c}$) and to the first term on the right side (at $C = c$, then averaged over c within $\bar{c}$) yields Eq. (12). □

Appendix A.5. Proof of Proposition 3

Fix a cell c with $n_c \geq 2$ and $i \in c$. By definition of the label count, $\sum_{y=1}^{K} m_{cy} H_i(y) = \sum_{j \in c} H_i(y_j) = H_i(y_i) + \sum_{j \neq i} H_i(y_j)$, and Eq. (22) gives $\sum_{j \neq i} H_i(y_j) = (n_c - 1)\hat{P}_i$. Hence

$$\tilde{P}_i = \frac{1}{n_c} \sum_y m_{cy} H_i(y) = \frac{H_i(y_i) + (n_c - 1)\hat{P}_i}{n_c}.$$

Substituting $H_i(y_i) = \hat{P}_i + \hat{R}_i$ gives $\tilde{P}_i = \hat{P}_i + n_c^{-1}\hat{R}_i$, and $\tilde{R}_i = H_i(y_i) - \tilde{P}_i = \hat{R}_i - n_c^{-1}\hat{R}_i = (n_c - 1)n_c^{-1}\hat{R}_i$. Averaging $\tilde{R}_i$ within cell c and reweighting by n_c across cells, as in the dataset-level definition of $\widehat{\Delta Z}$, gives $\widetilde{\Delta R} = N_*^{-1} \sum_c n_c \cdot \frac{n_c - 1}{n_c} \widehat{\Delta R}_c = N_*^{-1} \sum_c (n_c - 1)\widehat{\Delta R}_c$. □

Appendix A.6. Proof of Proposition 2

Fix $\phi = c$ throughout and write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$, both random through their dependence on (ϕ, Z). Let $U = (a_y - \mathbb{E}[a_y \mid c])_{y=1}^{K}$ and $V = (b_y - \mathbb{E}[b_y \mid c])_{y=1}^{K}$ be the corresponding mean-zero $\mathbb{R}^K$-valued random vectors conditional on $\phi = c$. The map $(U, V) \mapsto \mathbb{E}[\langle U, V \rangle \mid c] = \sum_y \operatorname{Cov}(a_y, b_y \mid c)$ is an inner product on the space of mean-square-integrable $\mathbb{R}^K$-valued random variables conditional on $\phi = c$, so by the Cauchy–Schwarz inequality applied to that inner product,

$$\Big|\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)\Big| \le \sqrt{\mathbb{E}[\|U\|^2 \mid c]}\,\sqrt{\mathbb{E}[\|V\|^2 \mid c]} = \sqrt{\textstyle\sum_y \mathrm{Var}(a_y \mid c)}\,\sqrt{\textstyle\sum_y \mathrm{Var}(b_y \mid c)}.$$

By the definition of ρ, the left side equals $|\rho|\sqrt{\sum_y \mathrm{Var}(a_y \mid c)}\sqrt{\sum_y \mathrm{Var}(b_y \mid c)}$ exactly, so this step alone is not yet a bound; two further steps supply one, at two different levels of conservatism. First, $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$ is itself a conditional expectation of $H_X(y)$ given more information than $\phi = c$ alone, so the law of total variance gives $\mathrm{Var}(H_X(y) \mid c) = \mathbb{E}[\mathrm{Var}(H_X(y) \mid \phi, Z) \mid c] + \mathrm{Var}(a_y \mid c) \ge \mathrm{Var}(a_y \mid c)$; likewise $b_y = \Pr(Y = y \mid \phi, Z)$ satisfies $\mathrm{Var}(b_y \mid c) \le \mathrm{Var}(\mathbb{1}\{Y = y\} \mid c) = p_y(c)(1 - p_y(c))$ by the same argument applied to $\mathbb{1}\{Y = y\}$ in place of $H_X(y)$. Both bounds hold termwise, so summing over y and applying the (monotone) square root to each sum separately gives the observable, per-cell bound

$$\sqrt{\textstyle\sum_y \mathrm{Var}(a_y \mid c)}\sqrt{\textstyle\sum_y \mathrm{Var}(b_y \mid c)} \le \sqrt{\textstyle\sum_y \mathrm{Var}(H_X(y) \mid c)}\sqrt{\textstyle\sum_y p_y(c)(1 - p_y(c))},$$

the second inequality of Eq. (15); this is a bound on a product of two sums, and does not simplify to a sum of per-label square roots, since Cauchy–Schwarz runs the other way for that rearrangement. Second, for Corollary 2, $a_y \in [-M, M]$ (a conditional expectation of a $[-M, M]$-valued variable) and $b_y \in [0, 1]$, so Popoviciu's inequality applied directly to a_y, b_y gives $\mathrm{Var}(a_y \mid c) \le M^2$ and $\mathrm{Var}(b_y \mid c) \le 1/4$ for every y, hence

$$\Big|\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)\Big| \le |\rho|\,\sqrt{KM^2}\,\sqrt{K/4} = \frac{|\rho|\,K\,M}{2}.$$

By Eq. (13), the left side of each display is exactly $|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]|$, giving Eq. (15) and Eq. (16). □

Appendix B. Influence function: estimand and derivation

The estimand under singleton exclusion.. Cells with $n_c = 1$ identify no within-cell counterfactual and are excluded, so the dataset-level statistic targets the *identified-subpopulation* quantity: writing π_c for the cell probabilities, the target of $\widehat{\Delta R}$ is the π-weighted mean $\theta = \sum_c \pi_c\, \Delta R_c / \sum_c \pi_c$ taken over cells that appear with at least two examples. At finite N the identified set is random; we treat the interval as conditional on identification and report the identified fraction alongside every estimate (99.0% of relations on VG150, 100% on CIFAR-100-LT), so the gap between the conditional and full-population targets is bounded by the reported coverage.

Derivation of Eq. (24).. Fix a cell c and let $\theta_c = \Delta R_c$ with $g_c(z) = \mathbb{E}[A(z, Z') \mid Z' \in c]$ the kernel projection. The classical Hájek projection of the order-two U-statistic $\widehat{\Delta R}_c$ is

$$\widehat{\Delta R}_c = \theta_c + \frac{2}{n_c}\sum_{i \in c}(g_c(Z_i) - \theta_c) + o_p(n_c^{-1/2}).$$

The dataset statistic weights cells by their example counts, $\widehat{\Delta R} = N_*^{-1} \sum_c n_c \widehat{\Delta R}_c$, and cell membership is itself random (multinomial over cells), so an example i contributes both through the within-cell projection and through the between-cell composition term $\theta_{C_i} - \theta$. Combining the two sources,

$$\widehat{\Delta R} - \theta = \frac{1}{N_*} \sum_i \Big[2(g_{C_i}(Z_i) - \theta_{C_i}) + (\theta_{C_i} - \theta) \Big] + o_p(N_*^{-1/2}).$$

Estimating $g_{C_i}(Z_i)$ by the pair-kernel mean $\bar{A}_i$, θ_c by $\widehat{\Delta R}_c$, and θ by $\widehat{\Delta R}$ gives the plug-in influence contribution $\widehat{\psi}_i = 2(\bar{A}_i - \widehat{\Delta R}_c) + (\widehat{\Delta R}_c - \widehat{\Delta R}) = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}$, which is Eq. (24) and has exactly zero sample mean by construction. Its empirical 95% coverage in a VG-like regime (image-clustered examples, Zipf cell sizes dominated by $n_c = 2$) is validated in Section 4.1.

Proof of Theorem 4.. The linearization above gives $\widehat{\Delta R} - \theta = N_*^{-1} \sum_i \psi_i + o_p(N_*^{-1/2})$ for the population influence contribution $\psi_i = 2(g_{C_i}(Z_i) - \theta_{C_i}) + (\theta_{C_i} - \theta)$, which has zero mean by construction and satisfies $|\psi_i| \le C(M)$ for an absolute constant $C(M)$ depending only on the score bound M in condition (i), since g_{C_i}, θ_{C_i}, and θ are themselves averages of the bounded kernel $A_{ij} \in [-2M, 2M]$. Because clusters are mutually independent (condition (ii)), $T_g = \sum_{i \in g} \psi_i$ for $g = 1, \ldots, G$ are independent, mean-zero random variables with $|T_g| \le |g|\, C(M)$. Condition (ii)'s finite third moment for cluster sizes and $\max_g |g| = o(\sqrt{G})$ give $\sum_g E|T_g|^3 \le C(M)^3 \sum_g \mathbb{E}|g|^3 = O(G)$ while $(\sum_g \mathrm{Var}(T_g))^{3/2} = \Theta(G^{3/2})$ under condition (iv), so Lyapunov's ratio $\sum_g E|T_g|^3 / (\sum_g \mathrm{Var}(T_g))^{3/2} \to 0$; the Lyapunov condition implies the Lindeberg condition, and the Lindeberg–Feller central limit theorem for triangular arrays of independent summands [47, 53] gives $(\sum_g T_g) / \sqrt{\sum_g \mathrm{Var}(T_g)} \xrightarrow{d} \mathcal{N}(0, 1)$. Condition (iii) makes $N_* / \sum_g \mathrm{Var}(T_g) \to 1/\sigma^2$ in probability by definition of σ^2, and the $o_p(N_*^{-1/2})$ remainder is asymptotically negligible after multiplying by $\sqrt{N_*}$, so $\sqrt{N_*}(\widehat{\Delta R} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ by Slutsky's theorem. For the variance estimator, $\widehat{\Delta R}_c \to_p \Delta R_c$ and $\widehat{\Delta R} \to_p \theta$ by the weak law of large numbers applied within each cell and across cells respectively (each is an average of a bounded kernel over a sample size diverging under (iii)), so $\widehat{\psi}_i \to_p \psi_i$ pointwise, and $\widehat{\sigma}^2$ – a sum of squares of cluster-aggregated $\widehat{\psi}_i$, normalized by N_* and G – is a continuous functional of these consistent plug-ins, hence $\widehat{\sigma}^2 \to_p \sigma^2$ by the continuous mapping theorem. □

Efficient computation.. The implementation does not instantiate all pairs. In addition to Eq. (22), let $G_c(y) = \sum_{j \in c} H_j(y)$ and $O_c = \sum_{j \in c} H_j(y_j)$. The pair-kernel mean for observation i is

$$\bar{A}_i = \frac{1}{2} \left[H_i(y_i) + \frac{O_c - H_i(y_i)}{n_c - 1} - P_i - \frac{G_c(y_i) - H_i(y_i)}{n_c - 1} \right].$$

Therefore both the point estimate and Hájek projection require only label counts, column sums, and matrix–vector products inside each cell. For image clusters g, the implemented variance is

$$\widehat{\mathrm{Var}}(\widehat{\Delta R}) = \frac{G}{G-1}\frac{1}{N_*^2}\sum_{g=1}^{G}\left(\sum_{i\in g}\widehat{\psi}_i\right)^2,$$

where G is the number of images and $\widehat{\psi}_i$ is Eq. (24). We use a normal critical value; with 27,955 image clusters, small-G corrections are immaterial.

Appendix C. Randomization algorithm

For each cell, sort examples once and index them $0, \ldots, n_c - 1$. Randomization b draws $o_c^{(b)}$ uniformly from this index set and maps label position r to $(r + o_c^{(b)}) \bmod n_c$. Independent offsets across cells define an element of the product of cyclic groups. The cell label histogram is unchanged, so $\sum_y m_{cy} H_i(y)$ in Eq. (22) is precomputed; only the assigned label lookup changes. One randomization is $O(N)$ after the $O(NK)$ setup. The plus-one Monte Carlo correction includes the observed assignment and prevents zero p-values.

Appendix D. Implementation and reproducibility

The NumPy implementation is in `wager/antisymmetric.py`. The primary routine `decompose_gain` validates probability matrices, forms score contrasts, excludes singleton cells, computes all three components, verifies the identity numerically, and returns per-instance transported/alignment values plus intervals. The Visual Genome driver is `experiments/run_sgg_wager.py`; controlled validation and sensitivity checks are in `experiments/antisymmetric_simulation.py` and `experiments/antisymmetric_ablations.py`. All random seeds are fixed. Unit tests cover the decomposition identity, quadratic covariance identity, cell-constant null, singleton reporting, cluster inference, randomization power, and log-score path, plus direct numerical checks of Propositions 3 and 1 (the latter against an explicit discrete population with exact expectations, independent of the estimator's own code path), Corollary 1's log-score covariance identity, Corollary 3's equivalence to the (unclustered) Diebold–Mariano statistic at a trivial audit feature, and Proposition 2's bound ordering (exact bias ≤ tight, per-cell bound ≤ crude, data-free bound) on a second explicit discrete population with a known omitted confounder. The covariance-identity cross-check quoted in Section 4.7 is reproduced by `experiments/cifar_covariance_check.py`; the calibration-matched protocol and the prior-matching experiment are `experiments/cifar_recalibration_control.py` and `experiments/vg_prior_consequence.py`. Table 1's evaluation of Corollary 2 is `experiments/sensitivity_bound_check.py`, which recomputes the crude bound directly from the committed `antisymmetric_ablations.json` rather than rerunning the decomposition; `experiments/sensitivity_analysis.py` computes Proposition 2's tighter, per-cell form from per-instance arrays (Section Appendix D.1).

Full-data VG predictors (Sections 4.3–4.4).. FREQ uses add-one smoothing over the 50 predicates within each training class pair, backing off to the subject-marginal and then the global predicate histogram for pairs unseen in training. The MLPs consume fixed random class embeddings (32 dimensions per class, seed 0, concatenated for subject and object) on standardized features and train with AdamW (learning rate 10^{-3}, weight decay 10^{-5}, batch size 4096): MLP-CLASS with hidden sizes $(256, 128)$ for 12 epochs, MLP-SPATIAL with $(256, 128)$ for 15 epochs, and MLP-SPATIAL+ with $(512, 256, 128)$ for 20 epochs – the "+" variant differs in both width and schedule, as noted in Section 4.2.

Data availability.. The repository ships the estimator, every driver script, and the cached results files that every printed number traces to (`results/*.json`). The cached per-relation probability arrays that the drivers consume (`sgg_dists.npz`, `vg_visual_models.npz`, `cifar_lt_results.npz`) are regenerable from the committed training scripts with fixed seeds and are archived alongside the code release, so exact reproduction does not require retraining.

Appendix D.1. Compute environment for Sections 4.6–4.7

Model training and feature extraction for both new experiments ran on a single NVIDIA T4 (15 GB) GPU. The WAGER decomposition itself is pure NumPy and requires no accelerator: it consumes the resulting cached probability arrays, as in Section 4.2. All Python code runs under a pinned environment (Python 3.13).

Matched-subsample models (Section 4.6).. The three compared predictors share a fixed, seeded (seed = 0) subsample of 100,000 of the 532,987 training relations, drawn without replacement uniformly at the relation level (not the image level); the full 229,605-relation test split is always used for evaluation. All three use the identical head and optimizer – AdamW, learning rate 10^{-3}, weight decay 10^{-5}, 15 epochs, batch size 4096, hidden sizes $(256, 128)$, on standardized features – so they differ only in their input features.

Subject/object crops are taken from the annotated boxes and CLIP-preprocessed (bicubic resize to 224×224 with CLIP's published normalization constants); a crop with no valid overlap with the image falls back to a blank crop rather than being dropped, so every relation contributes a feature row. CLIP runs in half precision with a batch size of 512 crops and is never fine-tuned. Two implementation details are worth recording. First, the pretrained OpenAI weights were trained with the QuickGELU activation, so the encoder must be instantiated as `ViT-B-32-quickgelu`; loading the same weights under the plain `ViT-B-32` configuration silently substitutes a standard GELU and degrades every embedding. Second, because an annotated object box participates in several relations, crops are deduplicated by (image, box) before encoding and gathered afterwards: the 659,210 crops implied by the audited relations collapse to 422,143 unique ones, removing 36.0% of the encoder work without changing any feature. Only the images referenced by those relations are retrieved, individually rather than through the two bulk archives; of the 71,990 required, three were unavailable upstream and fall back to a blank crop.

ResNet-32 triple (Section 4.7).. CIFAR-100-LT uses the standard exponential-profile construction of Cui et al. [11] at imbalance ratio 100: class c (zero-indexed, $0, \ldots, 99$) keeps $n_c = \lfloor 500 \cdot \mu^c \rfloor$ training images, where $\mu = 100^{-1/99}$, so class 0 keeps all 500 images and class 99 keeps 5; the balanced 10,000-image test split is untouched. All three models are a from-scratch, randomly initialized CIFAR ResNet-32 (three stages of five basic residual blocks, 16/32/64 channels) trained for 80 epochs with SGD (momentum 0.9, Nesterov, weight decay 2×10^{-4}), batch size 128, standard crop-and-flip augmentation, and a cosine learning-rate schedule (peak 0.1, five-epoch linear warmup). The only difference between the runs is the per-example loss weight. CE uses uniform weights throughout. CB applies the effective-number class-balanced weight $w_c \propto (1-\beta)/(1-\beta^{n_c})$ with $\beta = 0.9999$, renormalized to average to one across classes, from initialization. DRW trains with uniform weights until epoch 60 and switches to exactly the CB weights for the final twenty epochs [7]; no other hyperparameter changes at the switch. The robustness matrix of Table 8 repeats this recipe verbatim at seeds 1 and 2 (ratio 100) and at ratios 10 and 50 (seed 0), re-drawing the long-tailed subset with the configuration's own seed and initializing all three arms of a configuration identically (`experiments/colab_cifar_multiseed.py`). The post-hoc LA arm is not trained: it rescales the baseline's test probabilities by the inverse training class prior, $q_{\mathrm{LA}}(y \mid x) \propto q_{\mathrm{CE}}(y \mid x)/\pi_{\mathrm{train}}(y)$, renormalized.